The Pandemic Trilemma

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Abstract

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1 Introduction

A government confronting a pandemic faces three simultaneous objectives: protecting public health by minimizing excess mortality, preserving economic stability by minimizing the deviation of output from its potential, and maintaining fiscal sustainability by not destabilizing the level of public debt. This paper establishes that these three objectives are mutually incompatible due to the structural architecture of the pandemic environment—forming a *trilemma* in the precise economic sense of the Mundell-Fleming impossible trinity Mundell (1963): no feasible policy of a benevolent social planner can simultaneously achieve all three goals, regardless of the resources deployed. The impossibility arises from four properties of the pandemic environment that I formalize in Section 2 based on the economic theory and the existing literature. Viral prevalence of a deadly virus, once introduced exogenously, grows endogenously through human contact. As long as the virus is lethal e.g., there is no vaxine, the only instrument that protect the people is suppress transmission with instruments like—lockdowns, social distancing mandates, mobility restrictions—operate by reducing human interaction, which simultaneously contracts economic activity. Fiscal instruments can compensate for the resulting output loss, but they do so at the cost of public debt accumulation. And even in the absence of any discretionary fiscal response, the output contraction itself erodes the fiscal position through falling incomes, which reduce tax revenues and trigger mandatory transfer payments, generating debt independently of any policy choice.

Given this structure, the planner's decision is not whether to intervene fiscally but how to compose the intervention mainly choosing between two different fundamentally different transmission channels. Demand injection like for instance direct transfers, cash payments or expanded unemployment benefits sustains aggregate expenditure during containment, generating an immediate but transient output effect. Capacity preservation such as wage subsidies that maintain employment relationships, loan guarantees that prevent firm insolvency or equity injections that preserve sectoral viability reduces the persistence of the contraction by preventing the destruction of organizational capital, supplier networks, and job-specific human capital that would otherwise require costly reconstruction. These instruments carry different mechanisms, different fiscal costs, and different dynamic profiles. The optimal fiscal response therefore depends not only on its aggregate size but on its composition—and this composition should vary over the course of the pandemic as the state of the economy evolves transforming the social planner's decision into an intertemporal dynamic optimization problem.

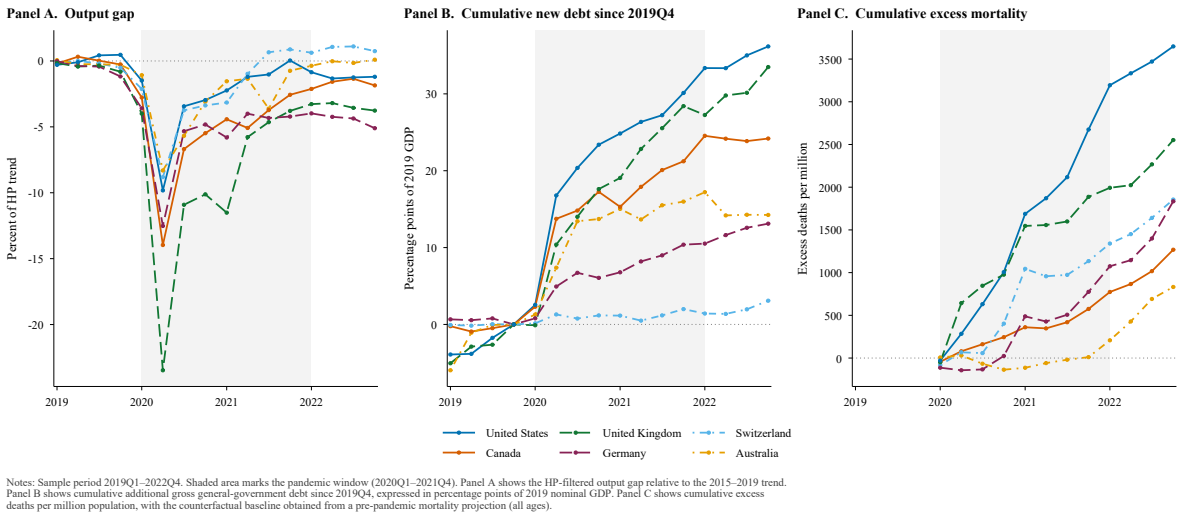

Figure 1: Pandemic Dynamics: Infection Prevalence, Excess Mortality, and Containment

A growing literature has formalized subsets of this problem. Alvarez et al. (2020) solve for the optimal lockdown trajectory of a planner trading off fatalities against output costs, while treating the economy as a linear production function with no fiscal dimension. Eichenbaum et al. (2021) endogenize the behavioral recession by embedding susceptible-infected-removed model (SIR) dynamics into a macroeconomic equilibrium, yet the government's only instrument is a Pigouvian containment tax—no fiscal compensation or debt dynamics enter the problem. Kaplan et al. (2020) come closest to the full trilemma: they integrate SIR dynamics into a heterogeneous-agent model with both lockdown and fiscal transfers, constructing a pandemic possibility frontier that maps policy combinations to the joint distribution of mortality and economic welfare losses. Their analysis establishes the binding health–economy trade-off and demonstrates that fiscal transfers reshape its distributional incidence. However, government debt enters as a residual accounting identity, not as an optimization target, and fiscal policy is modeled as undifferentiated lump-sum transfers rather than as a composition of instruments with distinct transmission mechanisms.[1]

Three gaps emerge from this body of work. First, no existing framework incorporates fiscal sustainability as a co-equal optimization target alongside health protection and output stabilization—the problem remains a dilemma rather than a trilemma. Second, the fiscal instruments available to the planner have not been disaggregated by their transmission mechanism which generate qualitatively different impact, dynamic profiles and carry different fiscal costs, losing the whole information about the optimal policy mix. Third, the instrument-specific structural parameters that govern the composition problem have not been estimated jointly under pandemic conditions using a transmission-mechanism-based disaggregation—existing cross-country evidence (Deb et al., 2021) benchmarks aggregate

[1]For a systematic comparison of modeling approaches in the pandemic-macro literature, see Table 19 in Appendix A.

fiscal effectiveness but not compositional optimality.

Closing these three gaps enables an analysis that no existing framework can deliver: given the containment trajectory that the prevailing epidemiological pressure necessitates, the planner can determine the optimal fiscal composition across demand injection and capacity preservation at each point in the pandemic, trace how this composition shifts as the state evolves, and quantify the output and debt consequences of deviating from it. This yields both a retrospective assessment of observed fiscal packages—identifying the excess debt attributable to suboptimal instrument choice—and a prospective benchmark for policymakers confronting future crises that share the pandemic's structural characteristics.

[CONTRIBUTION]

[MAIN FINDINGS]

[LITERATURE]

This paper builds up and contributes to three strands of literature. First, it extends the pandemic-macro modeling literature that integrates SIR dynamics with macroeconomic frameworks Atkeson (2020); Eichenbaum et al. (2021); Alvarez et al. (2021); Kaplan et al. (2020); Jones et al. (2021); Acemoglu et al. (2021) by incorporating fiscal sustainability as an explicit optimization objective and disaggregating fiscal instruments by transmission mechanism. Second, it contributes to the fiscal multiplier literature Blanchard and Perotti (2002); Ramey and Zubairy (2018); Romer and Romer (2010) by estimating instrument-specific multipliers under pandemic conditions, where Keynesian supply shocks Guerrieri et al. (2022) and voluntary demand reductions Goolsbee and Syverson (2021) render pre-crisis estimates unreliable. Third, it contributes to the empirical literature on cross-country fiscal responses to COVID-19 Deb et al. (2021); Chetty et al. (2020); Autor et al. (2022) by providing the first micro-level database with instrument-level granularity and source documentation across all OECD member states, enabling compositional analysis beyond aggregate fiscal impulse comparisons. [RELEVANCE]

The paper proceeds as follows. Section 2 establishes the four constitutive properties of the pandemic trilemma and its structural uniqueness relative to other crisis types. Section 3 develops the iLQR optimization framework. Section 4 describes the micro-level fiscal policy database. Section 4 presents the estimation strategy and structural parameter estimates. Section 4.4 reports the optimal composition trajectories and counterfactual exercises. Section 6 concludes.

2 The Pandemic Trilemma

This section formalizes the trilemma by identifying four constitutive properties of the pandemic environment—exogenous shock with endogenous lethality, contractionary suppression instruments, heterogeneous fiscal compensation channels, and endogenous fiscal feedback—and establishing their circular interdependence. Each property is developed with reference to the empirical literature and introduces the variables and parameters that enter the optimization framework in Section 3. I then apply the four properties as a diagnostic to alternative crisis types, establishing that the pandemic is the only configuration in which all four operate simultaneously within a tractable planning horizon.

2.1 Constitutive Properties

Property 1: Exogenous Shock with Endogenous Dynamics

A pandemic begins with the introduction of a novel pathogen into the population. I denote the pandemic pressure at time k by θ_k, which I refer to as the viral prevalence. The initial outbreak constitutes an exogenous shock—it originates outside the economic system and is not the result of endogenous market dynamics or policy failure. I represent this shock by a stochastic disturbance ε_k.

The defining characteristic of a pandemic shock is its endogenous growth dynamic: in the absence of intervention, each infected individual transmits the virus to more than one susceptible person, causing the number of infections to grow over time. I capture this by a persistence parameter $\rho_\theta > 1$ (similar to the logic of the reproduction rate during the pandemic), implying that viral prevalence grows exponentially if left unchecked (Atkeson, 2020; Alvarez et al., 2021; Eichenbaum et al., 2021).[2] It is the lethality of the pathogen that transforms this explosive dynamic into a policy imperative: a non-lethal pathogen with $\rho_\theta > 1$ would spread but impose no mortality cost of sufficient magnitude to warrant contractionary public health interventions. When the pathogen is lethal, unchecked exponential growth overwhelms the healthcare system and generates catastrophic mortality (Jones et al., 2021; Barro et al., 2020)—establishing the necessity of active policy intervention at the cost of economic contraction (Property 2).

This property distinguishes the pandemic shock qualitatively from an impulse shock—such as an earthquake or a one-off financial disruption—where the disturbance dissipates naturally over time ($\rho < 1$) and the system returns to equilibrium without active intervention (Jordà et al., 2022). These potential consequences are captured by the state variable:

[2] The linear specification abstracts from the susceptible-depletion mechanism of the full SIR framework, where the effective reproduction rate declines endogenously as the susceptible population shrinks. This linearization is appropriate for the model horizon (2020–2021), during which the pandemic evolved in discrete policy-interrupted waves rather than a single depletion arc, and variant emergence repeatedly replenished the effective susceptible pool. Residual nonlinearities—including behavioral adaptation and immunity waning—are absorbed by the stochastic disturbance ε_k.

excess mortality (d_k), defined as the deviation of the periodic death rate from its historical seasonal average encompassing both direct COVID-19 deaths and indirect deaths attributable to avoidance of care and healthcare system overload (Msemburi et al., 2023; Mathieu et al., 2020). I denote the parameter governing the transmission from infections to deaths by δ_θ. Crucially, the biological lag between infection and death implies that mortality in period $k+1$ is already determined by the current viral prevalence θ_k and thus predetermined from the planner's perspective. Consequently, the planner's intervention S_k—which suppresses transmission to θ_{k+1}—only impacts mortality starting from period $k+2$. This relationship is given by:

$$d_{k+2} = \delta_\theta \, \theta_{k+1}, \qquad \text{with} \quad \theta_{k+1} = \rho_\theta \theta_k + \epsilon_k \tag{1}$$

The interaction between the lethality parameter δ_θ and the transmission potential ρ_θ plays a decisive role. If $\delta_\theta = 0$—that is, if the virus were non-lethal—or if the pathogen were not transmissible ($\rho_\theta = 0$), there would be no reason for the planner to suppress θ_{k+1}. This logic is formally derived in Appendix C, which proves that the necessity of a trade-off-inducing policy ($S_k > 0$) depends strictly on the joint condition of positive lethality ($\delta_\theta > 0$) and active transmission ($\rho_\theta > 0$). Absent this conjunction, the suppression strategy becomes orthogonal to health outcomes, and no mortality costs or behavioral responses would arise.

Summary. Property 1 introduces the epidemiological state: the infection rate θ_k (self-reinforcing with $\rho_\theta > 1$), the mortality outcome d_k (driven by δ_θ), and the exogenous shock ε_k. Together, they establish that the planner must act to suppress viral prevalence, as the system is unstable without intervention.

Property 2: Suppression Instrument

Given the necessity of intervention established by Property 1, the planner's primary tool to suppress viral transmission during a pandemic is *non-pharmaceutical interventions* (NPIs): measures including, business closures, restrictions on gatherings, school closings, restrictions on internal movement as well as international travel controls and stay-at-home orders. I denote the intensity of these measures at time k by S_k, where $S_k = 0$ represents the absence of restrictions and $S_k = 1$ represents a complete lockdown in every aspect.

NPIs suppress viral prevalence by reducing human contact (Flaxman et al., 2020). I denote the effectiveness of this suppression by the parameter $\phi_S > 0$. Crucially, NPIs do not remove a fixed number of infections from the population; rather, they reduce the rate at which existing infections generate new ones. Each unit of suppression S_k reduces the transmission potential ρ_θ by the fraction ϕ_S, yielding an effective reproduction rate $\rho_\theta^{\text{eff}}(S_k) = \rho_\theta(1 - \phi_S S_k)$. Combining this with the mortality transmission from Property 1,

the full two-period channel from policy action to health outcome is:

$$d_{k+2} = \delta_\theta \cdot \underbrace{\left[\rho_\theta(1 - \phi_S S_k)\,\theta_k + \varepsilon_k\right]}_{\theta_{k+1}} \tag{2}$$

This equation encodes two features that are central to the planner's problem. First, the suppression effect is multiplicative in the current level of prevalence: at high θ_k, each unit of lockdown prevents more infections—and therefore more deaths—than at low θ_k. The marginal health benefit of suppression, $\partial d_{k+2}/\partial S_k = -\delta_\theta \rho_\theta \phi_S \theta_k$, is state-dependent and vanishes as the pandemic recedes (see Appendix C). Second, the planner does not control the infection level or the death rate directly but only the growth rate of the epidemic. When $\phi_S S_k$ is sufficient to push ρ_θ^{eff} below unity, infections decline; otherwise, they continue to grow despite the intervention. This threshold property—$\rho_\theta^{\text{eff}} < 1$ requires $S_k > \frac{1-1/\rho_\theta}{\phi_S}$—implies a minimum lockdown intensity below which suppression is ineffective at reversing the epidemic trajectory.

The central feature of Property 2 is that the channel through which NPIs suppress viral transmission—restricting human interaction—simultaneously reduces the volume of economic activity. Containment measures close businesses, idle labor, and curtail the flow of goods, services, and persons that constitutes aggregate economic output. This duality is not a matter of calibration or implementation; it is inherent to the nature of any instrument that operates through contact reduction.

To formalize this, I introduce the next state variable *output gap* (y_k): the percentage deviation of real GDP from its potential trend, where $y_k = 0$ represents normal economic conditions and $y_k < 0$ represents a recession. The contractionary effect of NPIs on output is governed by the parameter $\alpha_S > 0$. Every unit of suppression $S_k > 0$ therefore simultaneously implies some values of:

$$\text{Epidemiological benefit:} \qquad \Delta\theta = -\phi_S\, S_k < 0 \tag{3}$$

$$\text{Economic cost:} \qquad \Delta y = -\alpha_S\, S_k < 0 \tag{4}$$

The suppression instrument cannot control the epidemiological shock without simultaneously destabilizing the economic state. This dual impact is structural: it arises from the fact that α_S and ϕ_S are both strictly positive and operate through the same policy variable.

Beyond the direct contractionary effect of mandated containment, the pandemic generates an additional output cost through the behavioral response. When individuals observe rising death tolls, they voluntarily reduce economic activity independently of any government-mandated restrictions (Goolsbee and Syverson, 2021) making it a do or die strategy for the social planer as he can control it over the level of infection (or the lethal-

ity). I capture this channel through the parameter $\beta_d < 0$, which governs the output response to the mortality flow (d_k) and is commonly referred to as the *fear term*.

A natural objection to Property 2 is that alternative instruments, such as testing, might control the pandemic without suppressing economic activity, thereby dissolving the contractionary link. However, testing capacity constitutes a complement to, rather than a substitute for, NPIs (Eichenbaum et al., 2020). Critically, testing does not alter the biological parameters driving the pandemic: it reduces neither the inherent transmissibility (ρ_θ) nor the lethality (δ_θ) of the virus. Instead, it operates exclusively through the suppression channel S_k. By identifying infected agents, testing enables targeted isolation—which remains a form of NPI. Ideally, this improves the efficiency of the intervention, increasing the ratio of epidemiological benefit (ϕ_S) to economic cost (α_S) by replacing broad lockdowns with specific quarantines. However, testing cannot eliminate the contractionary effect ($\alpha_S > 0$) since even targeted isolation disrupts economic transactions. Moreover, the net effect on the efficiency ratio is ambiguous. Higher detection rates may intensify responses, though the net effect on the efficiency ratio is theoretically ambiguous.

Vaccination, by contrast, is the instrument that dissolves the trilemma by invalidating Property 2. It operates through both channels: reducing viral transmission among the immunized population (lowering ρ_θ) and reducing the lethality of breakthrough infections (lowering δ_θ)—which, as demonstrated in Appendix C, is the critical mechanism. Crucially, vaccination achieves both effects without operating through the suppression channel S_k—it requires no restriction of economic activity (Giordano et al., 2021), thereby eliminating the contractionary link that constitutes Property 2. However, effective vaccines were not widely available until mid-2021, and the critical vaccination thresholds required to sustainably decouple mortality from infection rates were not reached in most OECD economies until the end of 2021 (OECD, 2021). Consequently, for the main phase of the pandemic (Q1 2020–Q4 2021)—the period during which critical policy decisions were taken (Hale et al., 2021)—NPIs remained the only available suppression instrument, and were thus inherently contractionary.

Figure 5 illustrates with the OECD average the mechanism through which vaccination dissolves the trilemma. During Waves 1–3, infection prevalence and excess mortality move in parallel, and governments maintain elevated containment. With the onset of Wave 4 (Omicron, late 2021), the relationship breaks: infection prevalence reaches its pandemic peak, yet excess mortality declines sharply—reflecting the reduction in δ_θ achieved by widespread vaccination. Governments respond by withdrawing containment to near-zero levels despite record case counts. This decoupling is precisely the condition $\delta_\theta \to 0$ formalized above: once the fatality rate falls sufficiently, the rational to act vanishes, and the trilemma ceases to bind. This observation defines the planner's horizon: the trilemma is operative from Q1 2020 through Q4 2021—the period during which the joint condition of positive lethality ($\delta_\theta > 0$) and absence of effective vaccination necessitated the health–

economy–fiscal trade-off. Beyond the terminal date, the planner faces no further trilemma and deals exclusively with the fiscal legacy of the crisis.

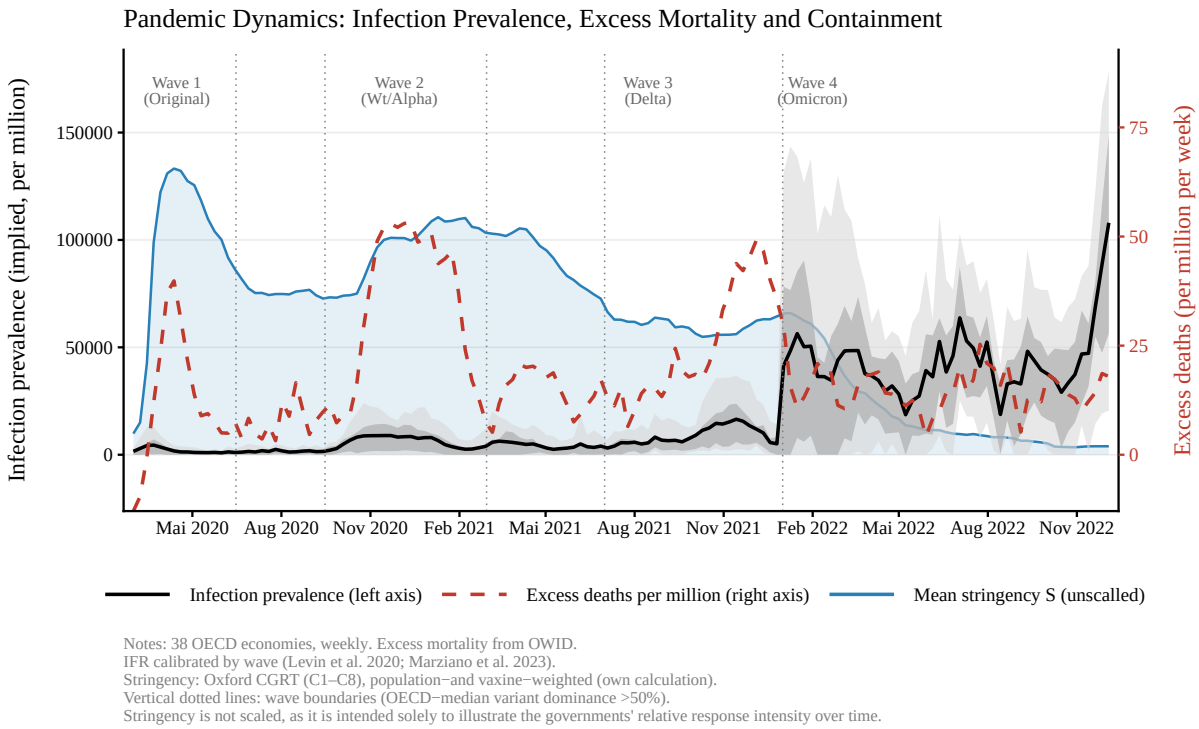

Figure 2: Pandemic Dynamics: Infection Prevalence, Excess Mortality, and Containment

Summary. Property 2 introduces the suppression instrument S_k, its epidemiological effectiveness ϕ_S, its direct economic cost α_S, the behavioral cost β_d driven by observed mortality, and the output gap y_k. It establishes that pandemic suppression is inherently contractionary: the planner cannot protect health without damaging the economy unless an effective vaccine is available.

Property 3: Fiscal Compensation Instruments

Given that pandemic suppression is inherently contractionary (Property 2), the social planner requires compensating instruments to stabilize the economic state. The pandemic damages the economy through two main channels. Containment and behavioral retrenchment directly reduce the level of output by restricting economic activity and suppressing expenditure. Simultaneously, the prolonged disruption erodes the economy's capacity to recover—through firm failures, dissolution of employment relationships, and disruption of supply chains—prolonging the contraction beyond the duration of the shock itself. To address these two channels, the social planner has to target these channels separately (level and persistence).

The first category, demand injection (F_k^{DI}) comprises measures that directly increase the flow of disposable income in the economy: stimulus checks (such as the U.S. Economic

Impact Payments under the CARES Act), direct transfers to households (such as the Australian JobSeeker supplement or the Japanese universal cash transfer of ¥100,000 per resident), and tax reductions that raise disposable income (such as the German temporary VAT reduction from 19% to 16% in H2 2020). These instruments transmit through the standard Keynesian multiplier channel and are widely used in depressed economic environment, shifting the output gap by $\alpha_{DI} > 0$ per unit deployed. [3]

However, applying these instrument to a pandemic environment is questionable on both theoretical and practical grounds. The pandemic recession is not a demand shock but a supply disruption induced by containment (Guerrieri et al., 2022): businesses are closed by government order, not by insufficient demand. Injecting purchasing power into an economy where the supply of goods and services is physically restricted cannot translate into expenditure at the usual rate (Chetty et al., 2020; Coibion et al., 2020). This raises the question whether α_{DI} under pandemic conditions retains a meaningful magnitude, is merely delayed, or vanishes entirely.

The second category, capacity preservation (F_k^{CP}), encompasses measures that maintain the existing productive structure of the economy: short-time work schemes (such as the Swiss *Kurzarbeitsentschädigung* or the UK Coronavirus Job Retention Scheme), payroll subsidies, and government-guaranteed loans (such as the French *Prêts Garantis par l'État* or the German KfW-Sonderprogramm). Unlike demand injection, capacity preservation targets the persistence of the shock rather than his level. When a lockdown closes a firm, CP does not restore its output—the firm remains closed regardless. What CP prevents is the destruction of firm-worker matches, supplier relationships, and organizational capital that would otherwise dissolve during the shutdown—precisely the amplification mechanism that Guerrieri et al. (2022) identify as propagating the initial sectoral shock. The preservation effect materializes only once containment eases and preserved capacity resumes production, making CP a structurally delayed instrument whose effectiveness depends on the depth of the contraction it is deployed against.

Combining the pandemic shocks from Property 2 with the fiscal instruments above, the output evolves according to:

$$y_{k+1} \;=\; \underbrace{\left(\rho_y + \psi\, S_k - \eta_p\, F_{k-1}^{CP}\right)}_{\equiv\, \rho_k^{\text{eff}}}\, y_k \;-\; \alpha_S\, S_k \;+\; \alpha_{DI}\, F_{k-1}^{DI} \;-\; \beta_d\, d_k. \tag{5}$$

Three features of equation (5) merit emphasis. First, the effective persistence $\rho_k^{\text{eff}} = \rho_y + \psi\, S_k - \eta_p\, F_{k-1}^{CP}$ is governed by two opposing forces. Containment raises persistence ($\psi > 0$) and capacity preservation lowers it ($\eta_p > 0$). Therefore, the net effect determines the speed of recovery—not the depth of the initial contraction, which is pinned down by the additive terms α_S and β_d.

[3] Pre-pandemic estimates place expenditure multipliers in a range of approximately 0.5 to 1.5 (Ramey and Zubairy, 2018; Blanchard and Perotti, 2002).

Second, CP is effective only when the economy is in recession. The marginal output effect of capacity preservation is $\eta_p\,|y_k|$: at $y_k = 0$, CP has no effect—there is no productive structure at risk. The deeper the contraction, the more matches and relationships face dissolution, and the larger the preservation benefit. This proportionality provides a formal rationale for the front-loading of CP programs observed across OECD economies during 2020, and implies that the optimal fiscal composition is state-dependent rather than fixed. The empirical counterpart is that corporate insolvencies across OECD economies fell by 17% on average in 2020 relative to 2019 despite the deepest recession since the Great Depression (Djankov and Zhang, 2021; Banerjee et al., 2021)—a pattern difficult to reconcile without the structural preservation channel formalized here.

Third, DI operates through a conventional level channel: $\alpha_{DI}\,F_{k-1}^{DI}$ shifts the output gap by a constant amount per unit deployed, independently of the economic state. This transmission mechanism is well-established for standard recessions where consumption channels are open (Ramey and Zubairy, 2018; Blanchard and Perotti, 2002). During a pandemic lockdown, however, containment restrictions constrain the expenditure margin, potentially attenuating the multiplier. Chetty et al. (2020) document that U.S. stimulus payments during strict containment were largely saved rather than spent, consistent with a blocked demand channel. The present framework nests this possibility: if $\alpha_{DI} = 0$, CP is the sole instrument through which fiscal resources translate into output recovery.

The preceding analysis might suggest a straightforward resolution: deploy CP until the persistence reduction offsets the lockdown amplification ($\eta_p\,F_{k-1}^{CP} \approx \psi\,S_k$), then inject demand into an economy whose productive structures are intact. However, this reasoning abstracts from the budgetary consequences of the fiscal package itself. Each instrument that stabilizes the output gap simultaneously augments the government's balance sheet—and the magnitude of this fiscal cost is heterogeneous across categories and in several dimensions not yet established in the literature. To formalize this constraint, I introduce the third state variable: *public debt* (b_k), defined as the deviation of the debt-to-GDP ratio from its pre-pandemic level.[4]

$$b_{k+1} = (1+r)\,b_k - \gamma_y\,y_k + \kappa^{CP}\,F_k^{CP} + \kappa^{DI}\,F_k^{DI} + c_H\,\theta_k \tag{6}$$

where r is the real interest rate, $\gamma_y > 0$ is the budget semi-elasticity (the automatic stabilizer channel formalized in Property 4), κ^{CP} and κ^{DI} are the budgetary cost parameters, and $c_H\,\theta_k$ captures pandemic-specific health expenditures.[5]

The budgetary cost parameter κ^{CP} is not a single number but a weighted average over

[4] I normalize by pre-pandemic GDP (2019) rather than current GDP and express it in real values. When debt is expressed relative to current output, a recession mechanically inflates the ratio through the denominator even absent any new borrowing. Normalizing by the pre-pandemic level isolates changes in the numerator—the actual stock of government liabilities—which is the object relevant to the planner's decision.

[5] See the discussion of health expenditures in the behavioral pandemic cost paragraph above.

two structurally distinct sub-categories. Direct capacity preservation—short-time work schemes, payroll subsidies, and direct grants—generates immediate budgetary outlays classified as above-the-line expenditure (International Monetary Fund, 2022). Loans and guarantees, classified as below-the-line, involve either a net fiscal cost well below unity (loans, where the government creates a financial asset offsetting the cash outflow) or near zero at deployment (guarantees, which generate no immediate outflow but create contingent liabilities whose realized cost depends on the default rate among supported borrowers). The effective raw effect of κ^{CP} therefore depends on the composition of the CP package and the realized default rate on contingent instruments.

The joint structure of equations (5) and (6) reveals that every unit of output stabilization carries a fiscal price—but this price differs across instruments and varies with the state of the economy. For capacity preservation, the marginal output gain from one additional unit of F^{CP} is $\eta_p\,|y_k|$—proportional to the depth of the recession. The fiscal cost per unit of output gap closed is therefore:

$$\pi_k^{CP} \;\equiv\; \frac{\kappa_i^{CP}}{\eta_p\,|y_k|} \tag{7}$$

This price is state-dependent: it falls with the depth of the recession as the preservation benefit grows, and rises toward infinity as the economy approaches full capacity ($y_k \to 0$). CP is cheap when it is most needed and prohibitively expensive when it is not—a property that provides a natural exit criterion for capacity preservation programs. For demand injection, the analogous price is:

$$\pi^{DI} \;\equiv\; \frac{\kappa^{DI}}{\alpha_{DI}} \tag{8}$$

This price is state-independent—it does not vary with S_k, y_k, or any other variable in the system. CP dominates DI whenever the recession is sufficiently deep that $\pi_k^{CP} < \pi^{DI}$, which yields the threshold condition:

$$|y_k| \;>\; \frac{\alpha_{DI}\,\kappa_i^{CP}}{\eta_p\,\kappa^{DI}} \;\equiv\; y^* \tag{9}$$

Below this threshold, the constant-return DI channel is more cost-effective per unit of debt. Above it, CP delivers more output recovery per fiscal dollar because the preservation benefit scales with the severity of the crisis. The model therefore predicts that the optimal fiscal composition—in terms of both effectiveness and cost—is not fixed but evolves endogenously with the state of the economy, shifting from CP-dominant during the acute contraction to DI-dominant during the recovery. This composition at each stage of the pandemic depends on the current state, on the anticipated trajectory of containment and infections, and on the cumulative fiscal legacy of past decisions—translating the problem into a genuinely dynamic one.

Summary. Property 3 introduces the fiscal compensation instruments—demand injection (F_k^{DI}) and capacity preservation (F_k^{CP}), disaggregated based on their transmission channels—and the public debt state (b_k). The two instruments operate through fundamentally different channels: DI shifts the output level through a state-independent multiplier α_{DI}, while CP reduces the effective persistence of the shock through $\eta_p\,|y_k|$—a benefit proportional to the depth of the recession. Each instrument that stabilizes output simultaneously augments the government's balance sheet, but the fiscal price per unit of stabilization differs across instruments and varies with the pandemic state. These state-dependent prices render the optimal composition and timing of the fiscal response an inherently dynamic problem.

Property 4: Endogenous Fiscal Feedback via Automatic Stabilizers

When NPIs contract economic output (Property 2), the resulting negative output gap triggers two simultaneous and automatic fiscal effects. On the revenue side, the tax base shrinks, reducing government receipts below their pre-pandemic trajectory. On the expenditure side, mandatory transfer programs such as unemployment insurance, social assistance, and housing benefits expand as more individuals satisfy eligibility criteria. The parameter $\gamma_y > 0$ in the debt transition (6) captures the aggregate sensitivity of the budget to the output gap, encompassing both channels in a single semi-elasticity.[6]

This term operates symmetrically. A negative output gap ($y_k < 0$) widens the deficit through lower revenues and higher mandatory expenditures; a narrowing gap ($y_k \to 0$) reverses both effects, reducing the pace of debt accumulation. The symmetry carries a direct implication for instrument choice: a fiscal instrument whose output effect is large relative to its fiscal cost generates a double dividend—it stabilizes output directly and attenuates the automatic fiscal deterioration indirectly, as the improved y_k feeds back through γ_y to slow debt accumulation.

The central contribution of this property is to close the trilemma by establishing that fiscal abstinence is not a feasible strategy (Fornaro and Wolf, 2020). Setting all discretionary instruments to zero ($F_k^{DI} = F_k^{CP} = 0$) in (6) yields:

$$b_{k+1} = (1 + r)\, b_k - \gamma_y\, y_k \tag{10}$$

Since $y_k < 0$ during containment, $-\gamma_y y_k > 0$: debt accumulates even without any discretionary fiscal expenditure. A lockdown without accompanying fiscal measures still erodes the fiscal position—the trilemma cannot be circumvented by fiscal abstinence. Combined with the necessity of suppression established in Property 1, this means that the planner's

[6]For OECD economies, estimates of the budget semi-elasticity range from approximately 0.35 to 0.65, with a cross-country average around 0.5 (Price et al., 2015). The parameter is structurally determined by the tax-to-GDP ratio and the coverage of automatic transfer programs.

decision is not whether to deploy fiscal support but which composition to deploy and when—returning the problem to the instrument choice formalized in Property 3.

Summary. Property 4 introduces the budget semi-elasticity γ_y and establishes that fiscal abstinence is an illusion: suppression passively erodes the fiscal position even absent discretionary intervention. The planner's decision is therefore not whether to intervene fiscally, but how to compose the intervention to minimize the joint debt footprint.

2.2 Summary

The four properties jointly characterize the structural environment of the pandemic trilemma. First, the system contains multiple simultaneously unstable dimensions: the epidemiological state is explosive without containment ($\rho_\theta > 1$); the output state becomes conditionally unstable under containment without capacity preservation ($\rho_y^{\text{eff}} \to 1$); and the debt state is unconditionally explosive, compounding any fiscal cost incurred during the crisis beyond the planning horizon ($(1 + r) > 1$). These instabilities jointly establish the necessity of planner intervention.

Second, every available instrument is structurally coupled across state dimensions. Containment suppresses infections but contracts output directly and through the behavioral response to mortality. Demand injection shifts the output level but augments debt. Capacity preservation reduces the persistence of the shock—a benefit proportional to the depth of the recession—but carries uncertain fiscal costs that depend on the composition of the package and the realized default rate on contingent instruments. No instrument operates on a single dimension, and mortality is not directly controllable, requiring a two-period transmission through the suppression channel.

Third, the marginal costs and benefits of each instrument are state-dependent. The epidemiological benefit of containment vanishes as infections recede. The output effect of capacity preservation scales with $|y_k|$, implying that CP is effective only when the economy is in recession and economically prohibitive once the output gap closes. The fiscal price of CP falls with the depth of the recession, while the price of DI is constant. The optimal instrument composition therefore evolves endogenously with the state of the economy.

Fourth, the system is inescapably constrained: every response to one dimension of the crisis worsens at least one other dimension, and fiscal abstinence worsens the fiscal dimension regardless through the automatic stabilizer channel (γ_y). The trilemma cannot be circumvented by inaction.

2.2.1 Structural Uniqueness

To establish that the pandemic trilemma represents a structurally distinct problem, I apply the four constitutive properties as a diagnostic framework to four alternative crisis types: armed conflict, natural disasters, financial crises, and climate change.[7] No other crisis type satisfies all four properties simultaneously. Armed conflict and natural disasters fail Property 2: the instruments that address the threat—military mobilization, reconstruction spending—are expansionary rather than contractionary. Financial crises fail Property 1: the shock is endogenous, and the primary interventions work with rather than against economic stabilization. Climate change approximates all four properties in principle, but differs in time scale (decades versus quarters), observability of the health outcome (diffuse versus immediate), and terminal condition: while technological decarbonization can in principle dissolve the climate trilemma—analogous to vaccination dissolving the pandemic trilemma—this resolution operates on a multi-decadal horizon with no guaranteed terminal date. The pandemic is therefore the only crisis type in which all four properties and their interdependence are simultaneously operative within a tractable planning horizon providing a unique situation.

3 The Planner's Problem

The four properties established in Section 2.1 together define the structural environment of the pandemic trilemma: the instability of the three state dimensions, the cross-dimensional coupling of every available instrument, the state-dependence of marginal costs and benefits, and the infeasibility of fiscal abstinence. This section assembles these elements into the formal decision problem faced by a social planner.

The state vector is $\mathbf{x}_k = (\theta_k,\, d_k,\, y_k,\, b_k)'$, collecting the epidemiological prevalence, observed mortality, output gap, and public debt defined in Section 2. The control vector is $\mathbf{u}_k = (S_k,\, F_k^{CP},\, F_k^{DI})'$, combining the containment instrument with the two fiscal instruments. The state evolves according to the following transition system:

$$\theta_{k+1} = \rho_\theta \left(1 - \phi_S S_k\right) \theta_k + \varepsilon_k \tag{11}$$

$$d_{k+1} = \delta_\theta \, \theta_k \tag{12}$$

$$y_{k+1} = \left(\rho_y + \psi \, S_k - \eta_p \, F_{k-1}^{CP}\right) y_k - \alpha_S \, S_k + \alpha_{DI} \, F_{k-1}^{DI} - \beta_d \, d_k \tag{13}$$

$$b_{k+1} = (1 + r) \, b_k - \gamma_y \, y_k + \kappa^{CP} \, F_k^{CP} + \kappa^{DI} \, F_k^{DI} + c_H \, \theta_k \tag{14}$$

The planner chooses the control sequence $\{\mathbf{u}_k\}_{k=0}^{N-1}$ to minimize a weighted quadratic loss

[7] See Appendix **??** for the complete diagnostic.

over the planning horizon: [8]

$$\min_{\{\mathbf{u}_k\}_{k=0}^{N-1}} \mathbb{E}\left\{\tfrac{1}{2}\,\boldsymbol{x}_N'\,\boldsymbol{Q}_N\,\boldsymbol{x}_N \;+\; \sum_{k=0}^{N-1}\tfrac{1}{2}\left(\boldsymbol{x}_k'\,\boldsymbol{Q}_k\,\boldsymbol{x}_k \;+\; \boldsymbol{u}_k'\,\boldsymbol{R}_k\,\boldsymbol{u}_k\right)\right\} \tag{15}$$

subject to (11)–(14) and the non-negativity constraints $S_k, F_k^{CP}, F_k^{DI} \geq 0$. The weighting matrices take the diagonal form $\boldsymbol{Q}_k = \beta^k \cdot \mathrm{diag}(0, w_d, w_y, w_b)$ and $\boldsymbol{R}_k = \beta^k \cdot \mathrm{diag}(r_S, r_{CP}, r_{DI})$, with terminal matrix $\boldsymbol{Q}_N = \beta^{N+1} \cdot \mathrm{diag}(0, 0, 0, W_b)$. The state weights $w_d, w_y, w_b > 0$ reflect the planner's preferences over mortality, output stabilization, and fiscal restraint; the control penalties $r_S, r_{CP}, r_{DI} > 0$ regularize the deployment of each instrument; and $\beta \in (0, 1)$ is the discount factor. The terminal weight W_b penalizes debt exclusively: output gaps close endogenously as containment is lifted, and mortality costs are irreversible and already penalized along the trajectory through w_d, but debt compounds at rate $(1 + r)$ beyond the planning horizon and constrains the fiscal response to subsequent shocks.[9]

The transition equations are specified as reduced-form relationships governing the evolution of the state vector, rather than equilibrium conditions derived from household and firm optimization. This places the model in the tradition of optimal policy analysis with reduced-form constraints—an approach with established precedents in the pandemic-macro literature (Alvarez et al., 2020; Acemoglu et al., 2021). The alternative—a fully microfounded DSGE framework—would derive the output dynamics from household and firm optimization, as in Eichenbaum et al. (2021) and Kaplan et al. (2020). However, the output response to containment in such models depends sensitively on functional form assumptions, household heterogeneity, and market structure, none of which bear directly on the fiscal composition question that this paper addresses. Section **??** discusses the relationship to this macro-modeling literature in more detail.

This choice of specification enables a decomposition that has not been formulated in the macroeconomic literature on pandemic policy: the separation of capacity preservation from demand injection on the basis of their transmission mechanism—level versus persistence—rather than their budgetary accounting. Guerrieri et al. (2022) identify firm exit and job destruction as propagation mechanisms that amplify the initial sectoral shock, providing the theoretical foundation for the persistence channel, but do not formalize capacity preservation as a distinct fiscal instrument operating through this channel. Deb et al. (2021) document empirically that emergency lifeline measures are more effective under strict containment while demand-support measures dominate during

[8] The quadratic objective (15) adopts a Tinbergen-style formulation that takes the multi-objective nature of pandemic policy as given, rather than deriving it from a representative agent's preferences—interpretable as a second-order approximation to such a welfare function (Woodford, 2003).

[9] This abstracts from the distributional dimension emphasized by Kaplan et al. (2020). The present framework addresses a logically prior question: what is the optimal *aggregate* fiscal composition? The distributional incidence of a given composition is a complementary but distinct problem.

reopening, but their classification follows budgetary accounting conventions—above-the-line versus below-the-line—rather than the economic transmission channel.[10] Gourinchas et al. (2021) quantify the firm-failure channel at the micro level, showing that fiscal support reduced SME failure rates by approximately 4.7 percentage points through liquidity preservation rather than demand stimulation—evidence for the persistence mechanism at the firm level that the present framework formalizes at the macro level. Together, the existing literature and the formalized framework motivate the central research question of this paper: *What is the optimal composition of fiscal support between capacity preservation and demand injection over the course of a pandemic, given the containment trajectory and epidemiological state, and what are the costs—in foregone output stabilization and excess debt—of deviating from this optimum?*

To answer this research question, the remainder of the paper proceeds the following steps. In Section 4 I estimate the structural parameters governing the transmission channels—ρ_y, ψ, η_p, α_S, α_{DI}, and β_d—from the cross-country panel, identifying the empirical strength and timing of each channel and providing estimates of the effectiveness and fiscal cost of each instrument. Section **??** calibrates the model using these estimates together with country-specific fixed effects and the epidemiological parameters (ρ_θ, δ_θ, ϕ_S) obtained from the data. Section **??** uses the calibrated model to solve the planner's minimization problem, constructing an optimal trajectory at the aggregate level and running scenario and country-specific counterfactuals: holding each country's observed pandemic conditions fixed—its containment trajectory, its epidemiological state, and the behavioral response to mortality—to answer which fiscal composition would have minimized the planner's loss given his pandemic conditions they faced and how far each country's realized policy departed from that conditional optimum.

Although the empirical analysis identifies how containment (S_k) and observed mortality (d_k) affect output, the paper does not model the epidemiological dynamics that generate these variables. I therefore fix each country's observed containment trajectory $\{S_k^{\text{obs}}\}$ and the induced epidemiological states $\{\theta_k^{\text{obs}}, d_k^{\text{obs}}\}$ throughout the optimization, and solve only for the fiscal controls. Two considerations motivate this restriction.

First, the epidemiological transition (11) depends on biological parameters (ρ_θ, δ_θ, ϕ_S) that lie outside the domain of fiscal policy analysis and that the present framework does not claim to model rigorously. Conditioning on observed containment and the resulting epidemiological trajectory keeps the analysis within the scope of this paper.[11]

Second, this restriction reflects the institutional reality of pandemic policymaking

[10] These categories cut across the distinction drawn here: above-the-line spending includes both demand-oriented measures (transfers, tax cuts) and capacity-preserving measures (short-time work, direct grants), while below-the-line spending captures part but not all of the capacity preservation category (loans and guarantees, but not payroll subsidies).

[11] Although the epidemiological dynamics are not targeted in the calibration, I use the fit of simulated θ_k and d_k against their observed counterparts as a non-targeted moment, providing independent evidence on model validity. See Section **??** for details.

rather than a modeling compromise. Governments during the pandemic did not choose containment and fiscal composition jointly: public health authorities set S_k based on epidemiological considerations, treating the protection of life as the binding priority, and finance ministries subsequently chose fiscal instruments to compensate for the economic consequences. The social planner in the present framework therefore faces the same sequential decision structure as real-world governments—taking containment as given and optimizing over the fiscal margin. The resulting problem is both policy-relevant and considerably less studied than optimal containment policy.

4 Empirics

4.1 Data and Measurement

The analysis draws on a quarterly panel dataset covering all 38 OECD member states. The underlying data span Q1.2015 through Q4.2024, though the estimation sample is substantially narrower. Table 1 summarizes all variables and sources.

Table 1: Variables and Data Sources

Variable	**Description**	**Source**
Dependent variables		
y_{ik}	Output gap: % deviation of real GDP from HP-filtered trend (2015–2019, $\lambda = 1600$)	OECD (2024)
b_{ik}	Quarterly change in real public debt, normalized by 2019 GDP (pp)	OECD (2024)
Policy variables		
S_{ik}	Stringency Index, population-weighted (0–100)	Own calc. (APX); Hale et al. (2021)
F_{ik}^{CP}	Capacity preservation (% of 2019 GDP)	Own database (APX)
F_{ik}^{DI}	Demand injection (% of 2019 GDP)	Own database (APX)
F_{ik}^{H}	Health spending (% of 2019 GDP)	Own database (APX)
Epidemiological variables		
d_{ik}	Excess mortality (P-score)	Msemburi et al. (2023); Mathieu et al. (2020)
θ_{ik}	Infection prevalence, estimated from excess mortality via country-specific IFRs	Own calc. (APX); Levin et al. (2020)
Vax_{ik}	Cumulative vaccination doses per capita	Mathieu et al. (2021)

Notes: All fiscal variables normalized by 2019 GDP. 2,301 individual measures classified by transmission channel and aggregated to country-quarter level. See Data Appendix for the classification protocol.

The output gap y_{ik} is measured as the percentage deviation of quarterly real GDP from its Hodrick–Prescott-filtered trend, estimated exclusively on pre-pandemic data (2015–2019, $\lambda = 1600$) and projected forward, yielding a country-specific counterfactual that is comparable across countries in percentage-point terms.[12] The debt variable b_{ik} is the

[12] Applying the HP filter to the full sample including 2020–2021 would pull the estimated trend down-

quarterly first difference of real public debt, normalized by 2019 GDP rather than contemporaneous GDP to isolate actual debt accumulation from denominator effects: a recession mechanically inflates the conventional debt-to-GDP ratio even without new borrowing. The fixed denominator ensures that b_{ik} is comparable across countries and over time in percentage-point terms.[13]

The fiscal variables F_{ik}^{CP}, F_{ik}^{DI}, and F_{ik}^{H} are constructed from a novel micro-level database of 2,301 individual policy measures across all 38 OECD member states, building on and substantially extending an initial draft by Lacey et al. (2021). The database records each measure at daily frequency with its announcement date, description, primary source, and fiscal cost (in local currency and as % of 2019 GDP). The classification distinguishes instruments by their primary economic transmission mechanism rather than by recipient or administrative category—the operationalization required by the theoretical framework in Section 2. Table 2 details the composition of each channel. The full classification protocol and a high-frequency descriptive analysis for all 38 OECD countries are documented in the Data Appendix (Appendix **??**).

Table 2: Fiscal Instrument Classification

Channel	**Instruments**	**Measures**
Capacity Preservation (F^{CP})		
Above-the-line	Short-time work, payroll subsidies, direct grants	948
Below-the-line	Government loans, credit guarantees	400
Demand Injection (F^{DI})		
Transfers	Cash transfers, unemployment top-ups, ad-hoc benefits	346
Tax measures	Income tax relief, VAT reductions, tax credits	162
Demand stimulus	Infrastructure, green investment, tourism support	38
Health (F^{H})		
	Medical procurement, hospital capacity, vaccine rollout	232
Excluded: Tax deferrals (IMF Cat. 3)		154
Excluded: Unclassified and excluded policy codes		21
Total		**2,301**

Notes: Classification by primary transmission mechanism, not by recipient or administrative category. Below-the-line CP adjusted to 35% estimated take-up rate in the debt equation (25%/50% robustness). Tax deferrals excluded as realized fiscal cost approaches zero. Full protocol in Data Appendix (Appendix **??**).

Excess mortality d_{ik}—the deviation of all-cause deaths from a projected seasonal baseline—is the preferred measure of pandemic health impact. Unlike reported COVID-19 deaths, which depend on testing capacity, health infrastructure and cause-of-death attribution

ward, mechanically reducing the measured output gap. The pre-pandemic-only estimation ensures that counterfactual potential output reflects the economy's trajectory absent the pandemic shock.

[13]The real debt series deflates nominal general government gross debt by the GDP deflator. Robustness checks using nominal debt and controlling for the inflation rate yield qualitatively identical results (Section 7).

practices that varied widely across countries, excess mortality captures both direct viral fatalities and indirect deaths from healthcare system overload, deferred care, and resource diversion (Msemburi et al., 2023). It is recorded with coverage across all OECD countries regardless of testing infrastructure, making it comparable across countries and over time. Excess mortality also enters the output equation as the fear term d_{ik}, controlling for voluntary reductions in economic activity: Goolsbee and Syverson (2021) show that observed lethality—not case counts—was the primary driver of voluntary behavioral responses during the pandemic.

Reported case counts are an unreliable measure of true infection prevalence, as testing capacity and reporting standards varied systematically across countries and over time. The infection prevalence variable θ_{ik} is therefore not measured directly but recovered by inverting the excess mortality series using country-specific age-adjusted infection fatality rates (Levin et al., 2020; COVID-19 Forecasting Team, 2022a). This yields a cross-country comparable measure of pandemic pressure that is not contaminated by differential testing regimes. As the pandemic-related variables are available at daily or weekly frequency, all variables are aggregated to the quarterly level to match the frequency of the macroeconomic outcomes.

4.2 Identification

The estimation targets the causal effects of Capacity Preservation (F_k^{CP}) and Demand Injection (F_k^{DI}) on the output gap y_k conditional on the containment level S_k, and the fiscal cost of each instrument in terms of debt accumulation b_k based on observables.

Three features of the pandemic setting pose potential identification challenges. First, containment and fiscal support could both be endogenous to pandemic severity: countries experiencing worse outbreaks imposed stricter lockdowns and deployed larger fiscal packages, generating a positive correlation between the treatment variables and the severity of the shock—both across countries and over time within countries.

Second, the two policy instruments—containment S_k and fiscal support F_k—are applied simultaneously to the same units, creating a double-treatment setting in which isolating the marginal effect of either instrument requires conditioning on the other. Third, treatment is continuous, time-varying, and global.

Yet the same setting that poses these challenges also provides the variation to overcome them. Critically, the within-country correlation between containment intensity and fiscal deployment over the sample period is below $r = 0.04$, [14] ruling out the concern

[14] Disaggregated pooled correlations: $r(S, F^{CP}) = 0.061$ ($p = 0.26$), $r(S, F^{DI}) = 0.059$ ($p = 0.28$), $r(F^{CP}, F^{DI}) = 0.233$ ($p < 0.001$). After absorbing country and quarter fixed effects (the identifying variation): $r(S, F^{CP}) = -0.066$, $r(S, F^{DI}) = -0.087$, $r(F^{CP}, F^{DI}) = 0.082$, $r(S \times F^{CP}, S \times F^{DI}) = -0.010$. The near-zero TWFE-demeaned correlations confirm that the identifying variation in fiscal composition is orthogonal to containment intensity.

that stricter containment induces larger fiscal compensation. Countries that were structurally comparable along many dimensions and faced similar pandemic intensity chose fundamentally different strategies for supporting their economies: the cross-country coefficient of variation of quarterly fiscal deployment is 1.39, compared to a CV of 0.22 in containment responses measured by the population weighted Stringency-Index (Hale et al., 2021)—a variation ratio of 6.3×. While the containment response was initially largely synchronized—with a cross-country correlation of quarterly time series of $r = 0.90$ during the first wave (March–May 2020), declining to $r = 0.56$ by 2021—the fiscal composition of the response differed from the outset, with a cross-country co-movement of only $r = 0.32$ during the same first wave. This compositional heterogeneity reflects differences in government preferences and pre-existing institutional frameworks—factors determined long before the pandemic and absorbed by country fixed effects insofar as they correlate with economic outcomes. This near-orthogonality shifts the analytical focus from the aggregate size of the fiscal response to its composition, timing, and state-contingency (Section 2)—and to the question which strategy delivered more stabilization per unit of debt incurred. [15]

The estimating equation maps directly onto the output transition (13) from the planner's problem:

$$y_{ik} = \rho_y\, y_{i,k-1} + \psi\, S_{ik}\, y_{i,k-1} + \eta_p\, F^{CP}_{i,k-2}\, y_{i,k-1} + \alpha_S\, S_{ik} + \alpha_{DI}\, F^{DI}_{i,k-2} + \beta_d\, d_{ik} + \gamma_i + \delta_k + \varepsilon_{ik} \quad (16)$$

where γ_i and δ_k denote country and quarter fixed effects. Each coefficient has a structural interpretation. The three persistence channels enter multiplicatively with the lagged output gap: ρ_y captures baseline mean reversion, ψ the lockdown amplification identified in Property 2, and η_p the capacity preservation channel identified in Property 3. The latter is the central parameter of this paper—$\eta_p < 0$ would indicate that CP reduces the effective persistence ρ_k^{eff}, consistent with the structural preservation mechanism. Three additive terms capture the level channels: the direct output cost of containment (α_S), the demand injection multiplier (α_{DI}), and the behavioral response to observed mortality (β_d), the latter controlling for voluntary reductions in economic activity driven by the fear term discussed in Section **??**.

The lag structure of both fiscal instruments is not imposed but selected empirically by comparing specifications with F^j_k, F^j_{k-1}, and F^j_{k-2} for $j \in \{CP, DI\}$; the data select lag 2 for both instruments. This is consistent with the standard fiscal transmission lag documented in the literature (Blanchard and Perotti, 2002; Ramey and Zubairy, 2018) and with the pandemic-specific prediction that fiscal policy is ineffective while supply constraints bind (Guerrieri et al., 2022)—instruments authorized during active contain-

[15] Tables 25 and 26 in Appendix F report the full within-between variance decomposition and pairwise correlation matrix for the full data sample confirming these patterns.

ment translate into output only after restrictions ease. For CP, the two-quarter delay additionally reflects the economic logic of the persistence channel: capacity preserved during the shutdown materializes in output only upon reopening. Section 7 reports all lag specifications. As a complement to the static TWFE specification, Section **??** reports local projection impulse responses (Jordà and Taylor, 2025) to illustrate the dynamic transmission profile of each instrument over a four-quarter horizon.

F^{CP} enters in aggregate because above-the-line and below-the-line instruments operate through the same persistence channel at the macro level. Section 7 verifies this assumption by disaggregating F^{CP} into its sub-categories (short-time work, grants, loans, guarantees) and testing whether the persistence coefficient is homogeneous across them. The debt equation corresponds to the debt transition (14):

$$b_{ik} = \gamma_y \, y_{ik} + \kappa^{\text{above}} \, F_{ik}^{CP,\text{above}} + \kappa^{\text{below}} \, F_{ik}^{CP,\text{below}} + \kappa^{DI} \, F_{ik}^{DI} + \kappa_H \, F_{ik}^{H} + \mu_i + u_{ik} \tag{17}$$

where μ_i denotes country fixed effects. The debt equation uses contemporaneous fiscal flows F_{ik}^{CP} and F_{ik}^{DI}, while the output equation uses their two-quarter lagged values. This asymmetry reflects the distinction between the timing of fiscal *outlay*—which occurs when funds are disbursed and enters the government's balance sheet immediately—and the timing of the fiscal transmission to output, which operates with the two-quarter delay identified above.

Unlike the output equation, the debt specification includes country fixed effects only, without quarter fixed effects or a linear time trend in the main specification.[16] Two considerations motivate this choice. First, the endogeneity concern that motivates quarter fixed effects in the output equation—S_k responding to the common pandemic trajectory—does not apply here, as the debt equation does not contain containment intensity as a regressor. Second, quarter fixed effects would absorb the common temporal pattern of fiscal deployment—all OECD countries ramped up spending in early 2020 and wound down over 2021—eliminating the temporal covariation between fiscal flows and debt accumulation that, together with the cross-country variation in composition, identifies the cost parameters κ^{above}, κ^{below}, and κ^{DI}. This identification strategy follows Deb et al. (2021), who similarly omit time fixed effects when estimating the effects of fiscal announcements on industrial production in order to preserve the temporal variation in fiscal deployment that is essential for identifying instrument-specific effects. The inclusion of y_{ik} as a regressor absorbs the common temporal variation driven by the synchronized recession through the automatic stabilizer channel, making explicit time controls redundant. The remaining common time-varying factors—interest rates and denominator effects—are addressed by construction: sovereign borrowing costs were near zero throughout the sample period, and b_{ik} is normalized by pre-COVID GDP (see Section 4.1).

[16] See robustness for a specification with a linear and a non-parametric time trend.

The inclusion of y_{ik} as a regressor also separates the direct fiscal cost of each instrument from the indirect fiscal savings operating through the automatic stabilizer channel. Because fiscal instruments affect the output gap (equation 16), and the output gap in turn affects debt through automatic stabilizers, the coefficient γ_y captures the fiscal cost of the recession while the κ coefficients measure the direct fiscal cost of each instrument, net of its output stabilization channel. The total debt cost of CP is therefore $\kappa^{CP} + \gamma_y \cdot (\partial y / \partial F^{CP})$, where the second term is the self-financing component operating through the persistence channel identified in (16).

Unlike in the output equation, where above- and below-the-line CP operate through a common persistence channel, the debt equation disaggregates F^{CP} by accounting treatment. Above-the-line CP (short-time work, payroll subsidies) generates direct government expenditure; below-the-line CP (loan guarantees) creates contingent liabilities whose realized fiscal cost is substantially below face value. Estimating a single κ^{CP} would conflate these fundamentally different debt implications.[17] Health spending F^H_{ik} is included to capture pandemic-specific expenditures (medical procurement, hospital capacity) that generate direct fiscal costs but operate through the epidemiological state rather than through the output channel.

4.2.1 Threats to Identification

Causal interpretation of equations (16) and (17) requires $E[\varepsilon_{ik} \mid \cdot] = 0$ and $E[u_{ik} \mid \cdot] = 0$. The most immediate concern is endogeneity. Stringency S_k is a policy choice responding to epidemiological conditions, raising the possibility that unobserved pandemic severity drives both containment and economic outcomes simultaneously. Quarter fixed effects absorb the common pandemic cycle generating the co-movement of S_k across countries; the residual within-quarter variation reflects country-specific policy choices that exhibit the near-orthogonality to fiscal composition documented above. The identifying condition is not that S_k is exogenous in general, but that containment was chosen independently of fiscal composition—that the decision to lock down did not depend on whether the economy would be supported primarily through wage subsidies or through transfers. The two-quarter lag of both fiscal instruments in (16) rules out contemporaneous reverse causation by construction: F^{CP}_{k-2} and F^{DI}_{k-2} were deployed before y_{ik} was observed or observable, and the instruments were activated in response to the lockdown decision rather than to GDP data, which became available only with substantial publication delay.

Unemployment is not included as a control because it is an outcome of the same process—CP, particularly short-time work, directly prevents unemployment by design. Conditioning on it would absorb part of the treatment effect; indeed, the robustness analysis confirms that CP significantly reduces unemployment (Section 7).

[17] The baseline specification adjusts below-the-line guarantees to a 35% take-up rate following ECB and IMF Fiscal Monitor (2022) estimates. Robustness checks at 25% and 50% are reported in Section 7.

The population-weighted Stringency Index captures mandated restrictions but not the voluntary reduction in economic activity driven by perceived infection risk (Chetty et al., 2020; Goolsbee and Syverson, 2021). If this voluntary response correlates with fiscal composition—for instance, if populations in countries with stronger safety nets reduce activity less—the estimated effect of S_k is attenuated. The mortality-driven fear term d_{ik} absorbs this channel directly through the coefficient β_d. The voluntary response does not threaten identification of the persistence coefficient η_p either, since CP eligibility was typically tied to mandated revenue losses rather than voluntary demand reductions. Robustness specifications using Google Mobility data in place of S_k confirm that ψ and η_p are insensitive to the operationalization of containment (Section 7).

Vaccination, available from Q1.2021, dissolves the trilemma by reducing the health cost of economic activity and is a potential confounder if countries that vaccinated faster simultaneously adjusted fiscal composition. The baseline controls for the cumulative vaccination rate, and results are robust to restricting the sample to the pre-vaccination period Q1.2020–Q4.2020, when the trilemma was fully binding (Section 7).

The classification of fiscal measures into CP and DI involves judgment calls that could introduce measurement error. If classification errors correlate systematically with S_k or y_k, the estimated coefficients are attenuated. The classification protocol (Section 4.1) mitigates this through rules based on the primary transmission mechanism; robustness specifications under alternative classification rules confirm that the results are not driven by borderline cases (Section 7).

Two further issues are inherent to the cross-country panel design. The synchronized pandemic generates dependence across countries through trade and supply chains. Symmetric spillovers—global demand contractions affecting all countries equally—are absorbed by quarter fixed effects. Asymmetric spillovers, such as differential EU single market exposure that varies over the pandemic, are not fully absorbed by either country or quarter fixed effects and constitute a maintained limitation, partially addressed by the finding that controlling for trade openness does not affect the results (Section 7).

Under these conditions, equations (16) and (17) identify identify the structural parameters of the transmission system.

4.3 Descriptives

Table 3: Summary Statistics

	Mean (SD)	Median	Min	P25	P75	Max
Panel A: Quarterly observations ($N = 380$ country-quarters)						
Outcome variables						
Output gap, pp of potential GDP	−4.36 (4.77)	−3.74	−23.47	−6.34	−1.74	11.13
Change in public debt, pp of 2019 GDP	0.85 (2.48)	0.46	−7.46	−0.61	2.00	14.20
Policy instruments						
Containment stringency index (0–100)	40.19 (18.70)	39.80	0.00	28.24	54.45	85.66
Capacity preservation F^{CP}, % of 2019 GDP	1.31 (3.03)	0.12	0.00	0.00	1.09	25.42
Demand injection F^{DI}, % of 2019 GDP	0.26 (0.71)	0.00	0.00	0.00	0.14	8.32
Health expenditures F^{H}, % of 2019 GDP†	0.12 (0.38)	0.00	0.00	0.00	0.00	4.25
Epidemiological controls						
Excess mortality, P-score (%)	10.23 (14.87)	5.51	−13.36	0.57	15.30	87.21
Infection prevalence, % of population	1.18 (1.89)	0.41	0.00	0.10	1.43	11.45
Cum. vaccination rate, % of pop.	31.35 (33.04)	12.46	0.00	0.00	66.52	89.68
Panel B: Cross-country totals ($N = 38$ countries)						
Outcome variables						
Avg. output gap, pp of potential GDP	−4.36 (3.05)	−4.22	−10.58	−5.96	−2.58	5.99
Cum. change in public debt, pp of 2019 GDP	8.55 (5.44)	8.47	−2.21	5.31	12.31	20.64
Policy instruments						
Avg. containment stringency (0–100)	40.19 (6.22)	40.25	27.66	36.51	43.65	56.09
Cum. capacity preservation F^{CP}, % of 2019 GDP	13.11 (6.99)	11.02	0.88	8.01	17.86	29.79
Cum. demand injection F^{DI}, % of 2019 GDP	2.61 (2.54)	1.89	0.11	0.95	3.83	11.89
Cum. health expenditures F^{H}, % of 2019 GDP†	1.15 (1.24)	0.73	0.00	0.31	1.70	4.73
Epidemiological outcomes						
Cum. excess mortality, P-score (%)	102.25 (76.95)	88.74	−13.59	49.54	131.05	355.59
Cum. infection prevalence, % of pop.	11.82 (5.39)	10.88	3.33	8.13	15.11	31.28
Cum. vaccination rate, % of pop.	73.31 (9.47)	74.50	45.67	67.49	80.31	89.68

Notes: Standard deviations in parentheses. Panel A reports the distribution of quarterly country-level observations; Panel B reports the cross-country distribution of period averages (output gap, stringency) and cumulative totals (all other variables) over Q1 2020–Q2 2022. All variables defined in Section 4.1. †Health expenditures enter the debt equation only; excluded from the output equation. The gap between mean and median for F^{CP} and F^{DI} in Panel A reflects the temporal concentration of fiscal deployment, with $F^{CP} = 0$ in 143 country-quarters (37.6%). $S = 0$ in five country-quarters (Q2 2022 only). Infection prevalence enters the iLQR calibration (Section **??**), not the main specification.

The table 3 provides descriptive statistics for the key variables of interest at the country-quarter (Panel A) and country level (Panel B), respectively. The average OECD country experienced an output gap of −4.4 pp over the sample period, with the deepest contraction reaching −23.5 pp in a single country-quarter and the worst cumulative outcome at −10.6 pp. Cumulative debt accumulation averaged 8.6 pp of 2019 GDP, ranging from −2.2 pp (net debt reduction) to 20.6 pp—a ninefold range that reflects the high differences between countries. Containment stringency was relatively synchronized across countries:

the cross-country standard deviation of period-average stringency is 6.2, compared to a within-country quarterly standard deviation of 18.7. After absorbing the common pandemic cycle through quarter fixed effects, sufficient country-specific variation remains to identify the effects of containment—see Section 4.2 for the decomposition. Fiscal composition exhibits the opposite pattern. The quarterly distributions of both F^{CP} and F^{DI} are strongly right-skewed: the mean of F^{CP} (1.31% of GDP) exceeds the median (0.12%) by an order of magnitude, with $F^{CP} = 0$ in 37.6% of country-quarters, reflecting the concentrated deployment of capacity preservation during the acute containment phase of 2020. Cumulatively, CP (13.1% of 2019 GDP) dominated DI (2.6%) by a factor of five, but the cross-country coefficient of variation is substantially larger for DI—countries that deployed demand injection at all did so at very different scales. This compositional heterogeneity, combined with the near-orthogonality to containment intensity documented in Section 4.2, constitutes the identifying variation.

The epidemiological variables display comparably wide cross-country dispersion. Cumulative excess mortality ranges from −13.6 P-score points—reflecting the net reduction in non-COVID mortality from reduced traffic, elective surgeries, and respiratory infections during lockdowns—to 355.6 points, a cumulative mortality burden more than three times the projected baseline. Infection prevalence and vaccination rates follow similar patterns of cross-country divergence.

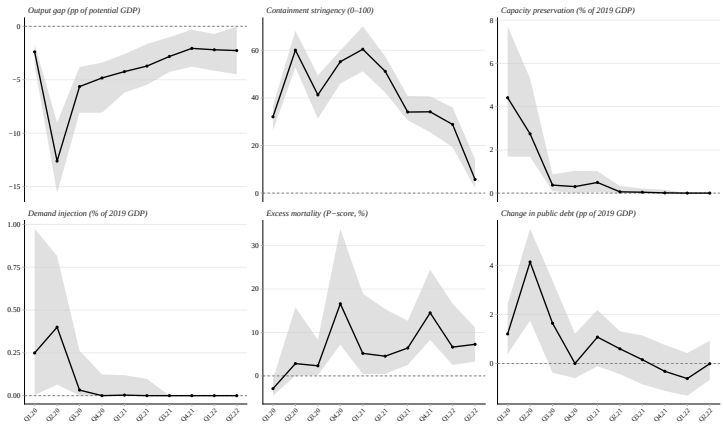

Figure 3: OECD Pandemic Trajectories by Quarter

Notes: Each panel plots the cross-country median (solid line) and interquartile range (P25–P75, shaded band) across 38 OECD countries per quarter, Q1 2020–Q2 2022. All variables are quarterly country-level observations as defined in Section 4.1. The narrow interquartile range for containment stringency relative to the wide bands for capacity preservation and demand injection reflects the compositional heterogeneity that drives identification.

4.4 Results

Table 4: Main Results: Output Gap Equation

	(1) Pooled OLS	(2) Country FE	(3) Country + Quarter FE
Persistence channels:			
$y_{i,k-1}$ (baseline ρ_y)	0.767***	0.174**	0.311***
	(0.080)	(0.067)	(0.085)
$S_{ik} \times y_{i,k-1}$ (lockdown ψ)	−0.004	0.000	0.002*
	(0.002)	(0.002)	(0.001)
$F^{CP}_{i,k-2} \times y_{i,k-1}$ (CP $-\eta_p$)	−0.024***	−0.005	−0.011***
	(0.007)	(0.003)	(0.003)
Level channels:			
S_{ik} (containment α_S)	−0.092***	−0.116***	−0.034***
	(0.019)	(0.013)	(0.012)
$F^{DI}_{i,k-2}$ (DI α_{DI})	0.999***	0.663***	0.185
	(0.198)	(0.109)	(0.122)
d_{ik} (fear β_d)	0.003	0.014	−0.021**
	(0.016)	(0.015)	(0.008)
Country FE		✓	✓
Quarter FE			✓
Observations	380	380	380
R^2	0.454	0.628	
Within R^2			0.281

Notes: Dependent variable: output gap y_{ik} (percentage points of potential GDP). Sample: 38 OECD countries over Q1 2020–Q2 2022 ($t \geq 5$, ten quarters per country). All columns use identical regressors; columns differ only in the fixed-effects structure. Standard errors clustered at the country level in parentheses, with small-sample adjustment. The progression across columns illustrates the identification strategy. The pooled OLS estimate of η_p (-0.024) is inflated by cross-country confounding between fiscal composition and country-specific recession depth. Country fixed effects absorb this confounding, and the coefficient collapses to -0.005 (insignificant), reflecting the loss of identification without temporal variation. Quarter fixed effects restore identification from within-country variation across quarters, yielding $\eta_p = -0.011$ ($p < 0.001$). The lockdown amplification parameter ψ follows a parallel pattern: negative in pooled OLS (sign reversal from omitted-variable bias), zero under country FE alone, and correctly signed and significant under the two-way specification. The level of F^{CP} is omitted because the theoretical framework predicts no contemporaneous level effect; a specification including F^{CP} as a level control is reported in Section 7. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Robustness: Persistence Channel η_p across Specifications

Specification	$\hat{\eta}_p$	SE	p-value		N
Panel A: Baseline & outlier exclusion					
Baseline	−0.011	(0.003)	0.000	***	380
No IRL	−0.011	(0.003)	0.000	***	370
No TUR	−0.011	(0.003)	0.000	***	370
No IRL + TUR	−0.010	(0.003)	0.001	***	360
Panel B: Alternative controls & measures					
$S_{ik}^{\max}$ instead of S_{ik}^{mean}	−0.011	(0.003)	0.000	***	380
+ Vaccination rate	−0.012	(0.003)	0.000	***	380
Fear: θ_{ik} (infection prevalence)	−0.011	(0.003)	0.000	***	380
Fear: $\bar{p}_{ik}$ (avg. P-score)	−0.011	(0.003)	0.000	***	380
+ Health expenditure (F^H)	−0.012	(0.003)	0.000	***	380
+ ICU occupancy	−0.009	(0.003)	0.010	**	194
+ $F_{i,k-2}^{CP}$ level	−0.021	(0.006)	0.001	***	380
Panel C: Sample variations					
Extended (Q4.2019–Q4.2022)	−0.013	(0.003)	0.000	***	456
Only 2020 (Q4.2019–Q4.2020)	−0.008	(0.003)	0.003	***	152
Shorter (Q1.2020–Q1.2022)	−0.010	(0.003)	0.001	***	342
Panel D: Sample splits					
High stringency	−0.015	(0.004)	0.001	***	190
Low stringency	+0.001	(0.008)	0.931		190
High income	−0.009	(0.003)	0.007	***	200
Low income	−0.017	(0.005)	0.001	***	200
High social safety nets	−0.010	(0.003)	0.004	***	190
Low social safety nets	−0.020	(0.010)	0.067	*	190
Significant at 1%: 17/20 at 5%: 18/20 at 10%: 19/20					

Notes: Each row re-estimates the full V3 specification $y_{ik} = \mu_i + \varepsilon_k + \rho\, y_{i,k-1} + \psi\, S_{ik} y_{i,k-1} - \eta_p\, F_{i,k-2}^{CP} y_{i,k-1} + \alpha_S\, S_{ik} + \alpha_{DI}\, F_{i,k-2}^{DI} + \beta_d\, d_{ik} + u_{ik}$ with the modification indicated. Standard errors clustered by country. The "$+F^{CP}$ level" row adds $F_{i,k-2}^{CP}$ as a level regressor; $\hat{\eta}_p$ doubles because the level absorbs part of the mean, leaving the interaction to capture steeper within-variation. The low-stringency subsample shows $\eta_p \approx 0$, consistent with the model prediction that the persistence channel requires an active lockdown shock: where S is low, output gap persistence is small and F^{CP} has no margin to preserve against. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Robustness: Demand Injection α_{DI} across Specifications

Specification	$\hat{\alpha}_{DI}$	SE	p-value		N
Panel A: Baseline & outlier exclusion					
Baseline	+0.185	(0.122)	0.138		380
No IRL	+0.138	(0.128)	0.287		370
No TUR	+0.199	(0.118)	0.102		370
No IRL + TUR	+0.148	(0.125)	0.246		360
Panel B: Alternative controls & measures					
$S_{ik}^{\max}$ instead of S_{ik}^{mean}	+0.222	(0.120)	0.074	*	380
+ Vaccination rate	+0.184	(0.116)	0.123		380
Fear: θ_{ik} (infection prevalence)	+0.206	(0.127)	0.114		380
Fear: $\bar{p}_{ik}$ (avg. P-score)	+0.186	(0.122)	0.137		380
+ Health expenditure (F^H)	+0.189	(0.124)	0.138		380
+ ICU occupancy	+0.136	(0.189)	0.480		194
+ $F_{i,k-2}^{CP}$ level	+0.215	(0.115)	0.070	*	380
Panel C: Lag structure					
$F_{i,k-1}^{DI}$ (lag 1)	−0.077	(0.108)	0.478		380
$F_{i,k-3}^{DI}$ (lag 3)	+0.021	(0.134)	0.876		380
Panel D: Sample variations					
Extended (Q4.2019–Q4.2022)	+0.153	(0.130)	0.244		456
Only 2020 (Q4.2019–Q4.2020)	+0.060	(0.248)	0.812		152
Shorter (Q1.2020–Q1.2022)	+0.200	(0.125)	0.119		342
Panel E: Sample splits					
High stringency	+0.221	(0.128)	0.102		190
Low stringency	+0.125	(0.103)	0.243		190
High income	+0.119	(0.157)	0.455		200
Low income	+0.180	(0.158)	0.267		200
High social safety nets	−0.082	(0.244)	0.739		190
Low social safety nets	+0.269	(0.133)	0.058	*	190
Significant at 1%: 0/22 at 5%: 0/22 at 10%: 3/22					

Notes: Each row re-estimates the full V3 specification with the modification indicated. α_{DI} is the coefficient on $F_{i,k-2}^{DI}$ (lag 2 unless otherwise noted). Standard errors clustered by country. The coefficient is consistently positive (+0.12 to +0.27) across 19 of 22 specifications but rarely reaches conventional significance, reflecting (i) 58% zero observations in F^{DI}, (ii) the two-quarter lag reducing effective variation, and (iii) small magnitudes (mean 0.26 pp GDP vs 1.31 for F^{CP}). Panel C confirms the lag 2 timing: lag 1 is negative and insignificant (too early), lag 3 attenuates to zero (too late), consistent with a disbursement-to-spending delay of approximately two quarters. The low-social-safety-net subsample — dominated by countries that relied more heavily on direct transfers (USA, CAN, CHL, AUS) — yields the strongest and only marginally significant effect (+0.269, $p = 0.058$). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Robustness: Stringency α_S across Specifications

Specification	$\hat{\alpha}_S$	SE	p-value		N
Panel A: Baseline & outlier exclusion					
Baseline	−0.034	(0.012)	0.006	***	380
No IRL	−0.032	(0.015)	0.040	**	370
No TUR	−0.031	(0.012)	0.016	**	370
No IRL + TUR	−0.025	(0.016)	0.123		360
Panel B: Alternative controls & measures					
$S_{ik}^{\max}$	−0.024	(0.009)	0.013	**	380
+ Vaccination rate	−0.036	(0.012)	0.006	***	380
Fear: θ_{ik}	−0.037	(0.012)	0.003	***	380
Fear: $\bar{p}_{ik}$	−0.034	(0.012)	0.006	***	380
+ Health expenditure (F^H)	−0.034	(0.012)	0.006	***	380
+ ICU occupancy	−0.028	(0.022)	0.210		194
+ $F_{i,k-2}^{CP}$ level	−0.033	(0.012)	0.008	***	380
Panel C: Sample variations					
Extended (Q4.2019–Q4.2022)	−0.033	(0.010)	0.001	***	456
Only 2020	−0.070	(0.021)	0.002	***	152
Shorter (Q1.2020–Q1.2022)	−0.033	(0.014)	0.019	**	342
Panel D: Sample splits					
High stringency	−0.031	(0.018)	0.113		190
Low stringency	−0.024	(0.018)	0.188		190
High income	−0.035	(0.015)	0.026	**	200
Low income	−0.031	(0.019)	0.118		200
High social safety nets	−0.001	(0.023)	0.977		190
Low social safety nets	−0.046	(0.013)	0.003	***	190
Significant at 1%: 8/20 at 5%: 13/20 at 10%: 13/20					

Notes: Each row re-estimates the full V3 specification with the modification indicated. α_S is the direct effect of the stringency index S_{ik} on the output gap. Standard errors clustered by country. The coefficient is consistently negative (−0.024 to −0.070) and significant in 13 of 20 specifications. The "$S^{\max}$" row uses the within-quarter maximum instead of the mean. The coefficient roughly doubles in the 2020-only subsample (−0.070), consistent with stringency having its strongest bite during the initial lockdown waves. The high-social-safety-net subsample shows $\alpha_S \approx 0$, suggesting that generous automatic stabilizers offset the direct output cost of lockdowns. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Robustness: Persistence ρ_y across Specifications

Specification	$\hat{\rho}_y$	SE	p-value		N
Panel A: Baseline & outlier exclusion					
Baseline	+0.311	(0.085)	0.001	***	380
No IRL	+0.270	(0.115)	0.024	**	370
No TUR	+0.273	(0.094)	0.006	***	370
No IRL + TUR	+0.198	(0.122)	0.114		360
Panel B: Alternative controls & measures					
$S_{ik}^{\max}$	+0.324	(0.090)	0.001	***	380
+ Vaccination rate	+0.310	(0.080)	0.000	***	380
Fear: θ_{ik}	+0.276	(0.091)	0.004	***	380
Fear: $\bar{p}_{ik}$	+0.309	(0.085)	0.001	***	380
+ Health expenditure (F^H)	+0.323	(0.089)	0.001	***	380
+ ICU occupancy	+0.026	(0.177)	0.884		194
+ $F_{i,k-2}^{CP}$ level	+0.329	(0.087)	0.001	***	380
Panel C: Sample variations					
Extended (Q4.2019–Q4.2022)	+0.521	(0.073)	0.000	***	456
Only 2020	−0.021	(0.107)	0.843		152
Shorter (Q1.2020–Q1.2022)	+0.169	(0.091)	0.073	*	342
Panel D: Sample splits					
High stringency	+0.418	(0.100)	0.001	***	190
Low stringency	−0.012	(0.128)	0.928		190
High income	+0.290	(0.136)	0.046	**	200
Low income	+0.422	(0.122)	0.003	***	200
High social safety nets	+0.182	(0.217)	0.412		190
Low social safety nets	+0.385	(0.085)	0.000	***	190
Significant at 1%: 12/20 at 5%: 14/20 at 10%: 15/20					

Notes: Each row re-estimates the full V3 specification with the modification indicated. ρ_y is the baseline persistence of the output gap (at $S = 0$ and $F^{CP} = 0$). Standard errors clustered by country. The coefficient is positive and significant in 15 of 20 specifications, with a baseline estimate of 0.311. Persistence rises in the extended sample (0.521) as recovery quarters strengthen the AR(1) pattern. The 2020-only subsample shows $\rho_y \approx 0$: with only four pandemic quarters, the output gap path is dominated by the lockdown shock and rebound rather than autoregressive dynamics. Low-stringency countries also show zero persistence, consistent with the model: where lockdowns were mild, output gaps were small and transitory. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Robustness: Lockdown Amplification ψ across Specifications

Specification	$\hat{\psi}$	SE	p-value		N
Panel A: Baseline & outlier exclusion					
Baseline	+0.002	(0.001)	0.075	*	380
No IRL	+0.003	(0.002)	0.093	*	370
No TUR	+0.003	(0.001)	0.073	*	370
No IRL + TUR	+0.004	(0.002)	0.053	*	360
Panel B: Alternative controls & measures					
$S_{ik}^{\max}$	+0.002	(0.001)	0.097	*	380
+ Vaccination rate	+0.002	(0.001)	0.052	*	380
Fear: θ_{ik}	+0.003	(0.001)	0.028	**	380
Fear: $\bar{p}_{ik}$	+0.002	(0.001)	0.068	*	380
+ Health expenditure (F^H)	+0.002	(0.001)	0.097	*	380
+ ICU occupancy	+0.005	(0.002)	0.044	**	194
+ $F_{i,k-2}^{CP}$ level	+0.002	(0.001)	0.055	*	380
Panel C: Sample variations					
Extended (Q4.2019–Q4.2022)	+0.001	(0.001)	0.613		456
Only 2020	+0.002	(0.002)	0.329		152
Shorter (Q1.2020–Q1.2022)	+0.004	(0.001)	0.009	***	342
Panel D: Sample splits					
High stringency	+0.002	(0.001)	0.194		190
Low stringency	+0.004	(0.003)	0.145		190
High income	+0.002	(0.002)	0.319		200
Low income	+0.001	(0.002)	0.467		200
High social safety nets	+0.005	(0.003)	0.122		190
Low social safety nets	+0.001	(0.001)	0.568		190
Significant at 1%: 1/20 at 5%: 3/20 at 10%: 12/20					

Notes: Each row re-estimates the full V3 specification with the modification indicated. ψ is the lockdown amplification parameter from the interaction $S_{ik} \times y_{i,k-1}$: higher stringency increases the persistence of the output gap. Standard errors clustered by country. The coefficient is consistently positive across all 20 specifications (+0.001 to +0.005) and significant at 10% in 12 of 20. The effect is identified most precisely in the core pandemic window (Q1.2020–Q1.2022: $\psi = 0.004$, $p < 0.01$) and attenuates in the extended sample where post-pandemic quarters dilute the lockdown-persistence nexus. Subsample splits lack power due to halved sample size, but all point estimates remain positive. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Robustness: Fear Term β_d across Specifications

Specification	$\hat{\beta}_d$	SE	p-value		N
Panel A: Baseline & outlier exclusion					
Baseline	−0.021	(0.008)	0.017	**	380
No IRL	−0.022	(0.008)	0.012	**	370
No TUR	−0.024	(0.008)	0.006	***	370
No IRL + TUR	−0.024	(0.008)	0.006	***	360
Panel B: Alternative controls & measures					
$S_{ik}^{\max}$	−0.023	(0.009)	0.015	**	380
+ Vaccination rate	−0.020	(0.008)	0.017	**	380
Fear: θ_{ik}	−0.008	(0.094)	0.931		380
Fear: $\bar{p}_{ik}$	−0.018	(0.008)	0.023	**	380
+ Health expenditure (F^H)	−0.021	(0.008)	0.018	**	380
+ ICU occupancy	−0.037	(0.023)	0.124		194
+ $F_{i,k-2}^{CP}$ level	−0.020	(0.008)	0.021	**	380
Panel C: Sample variations					
Extended (Q4.2019–Q4.2022)	−0.023	(0.008)	0.006	***	456
Only 2020	−0.016	(0.018)	0.382		152
Shorter (Q1.2020–Q1.2022)	−0.020	(0.009)	0.038	**	342
Panel D: Sample splits					
High stringency	−0.034	(0.014)	0.025	**	190
Low stringency	−0.007	(0.012)	0.578		190
High income	−0.038	(0.026)	0.165		200
Low income	−0.023	(0.008)	0.006	***	200
High social safety nets	−0.043	(0.022)	0.066	*	190
Low social safety nets	−0.012	(0.009)	0.182		190
Significant at 1%: 4/20 at 5%: 12/20 at 10%: 13/20					

Notes: Each row re-estimates the full V3 specification with the modification indicated. β_d captures the voluntary demand reduction from infection fear (P-score excess mortality). Standard errors clustered by country. The coefficient is negative and significant in 13 of 20 specifications. The "θ_{ik}" row replaces the P-score with infection prevalence — the coefficient collapses because θ is collinear with the quarter FE (common pandemic waves). The "$\bar{p}_{ik}$" row uses the historical-average rather than projection-based P-score, with similar results. The fear effect is concentrated in high-stringency and high-social-safety-net countries, where lockdowns made mortality salience a binding constraint on consumer behaviour beyond the mechanical lockdown effect. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 11 decomposes the variance of each regressor into between-country, within-country, and residual components after two-way fixed effects absorption. Two patterns are central to identification.

Table 11: Variance Decomposition of Regressors

	Total SD	Between SD	Within SD	R^2 absorbed (C+Q FE)
Output gap, y_{ik}	4.77	3.05	3.66	0.843
Output gap, $y_{i,k-1}$	4.72	2.79	3.81	0.827
Stringency, S_{ik}	18.70	6.22	17.63	0.756
Capacity preservation, $F^{CP}_{i,k-2}$	3.03	0.70	2.95	0.448
Demand injection, $F^{DI}_{i,k-2}$	0.71	0.25	0.66	0.246
Excess mortality, d_{ik}	14.87	7.70	12.73	0.433
$S_{ik} \cdot y_{i,k-1}$	240.77	122.80	207.10	0.725
$F^{CP}_{i,k-2} \cdot y_{i,k-1}$	38.14	10.39	36.70	0.428

Notes: Estimation sample (Q1 2020–Q2 2022, $N = 380$). R^2 absorbed reports the share of raw variance explained by country and quarter fixed effects jointly.

First, containment and the output gap are strongly synchronized across countries—fixed effects absorb 76% and 84% of their variance, respectively—reflecting the common epidemiological cycle. Fiscal deployment, by contrast, is substantially less synchronized: 55% of the variance in F^{CP}_{k-2} and 75% in F^{DI}_{k-2} remains as identifying variation after absorption of fixed effects. This asymmetry reflects the institutional separation between health authorities, which coordinated containment along the epidemiological cycle, and finance ministries, which deployed fiscal instruments along national administrative timetables.

Second, Table 12 reports the correlation matrix of the fixed-effects residuals. The correlation between containment and fiscal instruments is effectively zero (-0.014 with F^{CP}_{k-2}; -0.020 with F^{DI}_{k-2}), and the two fiscal instruments are only weakly correlated with each other (0.082). The non-negligible correlation between containment and excess mortality (0.182) reflects the well-known feedback from observed lethality to policy stringency; conditional on both S_{ik} and d_{ik}, the fiscal coefficients are identified from variation in instrument composition orthogonal to the severity of the crisis.

Table 12: Residual Correlations after Country and Quarter FE

	S_{ik}	$F^{CP}_{i,k-2}$	$F^{DI}_{i,k-2}$	d_{ik}
S_{ik}	1.000	−0.014	−0.020	0.182
$F^{CP}_{i,k-2}$	−0.014	1.000	0.082	0.015
$F^{DI}_{i,k-2}$	−0.020	0.082	1.000	−0.064
d_{ik}	0.182	0.015	−0.064	1.000

Notes: Correlations of residuals after two-way fixed effects demeaning. Estimation sample (Q1 2020–Q2 2022, $N = 380$).

Table 13: Main Results: Debt Equation

	(1) OLS	(2) Country FE	(3) Country FE — CP decomposition	(4)	(5)
		Pooled	Raw	Adj. 35%	Three-way
y_{ik} $[-\gamma_y]$	−0.147***	−0.219***	−0.215***	−0.215***	−0.210***
	(0.029)	(0.036)	(0.036)	(0.036)	(0.037)
F_{ik}^{CP} (pooled) $[\kappa^{CP}]$	0.186***	0.167**			
	(0.049)	(0.048)			
F^{CP} above-the-line $[\kappa^{\text{above}}]$			0.401*	0.401*	0.423*
			(0.176)	(0.176)	(0.174)
F^{CP} below-the-line $[\kappa^{\text{below}}]$			0.112*		
			(0.045)		
F^{CP} below-the-line (adj.) $[\kappa^{\text{below}}]$				0.321*	
				(0.127)	
F^{CP} loans $[\kappa^{\text{loans}}]$					0.352***
					(0.083)
F^{CP} guarantees (adj.) $[\kappa^{\text{guar}}]$					0.095
					(0.164)
$F_{i,k}^{DI}$ $[\kappa^{DI}]$	0.375	0.161	0.215	0.215	0.181
	(0.293)	(0.281)	(0.283)	(0.283)	(0.282)
Country FE		✓	✓	✓	✓
Guarantee adjustment	—	—	No	35%	35%
Observations	380	380	380	380	380
Within R^2		0.220	0.228	0.228	0.238

Notes: Dependent variable: Δb_{ik} (quarterly first difference of real public debt, pp of 2019 GDP). 38 OECD countries, Q1 2020–Q2 2022 (380 observations). Standard errors clustered by country in parentheses. All fiscal variables in % of 2019 GDP. Column (1): pooled OLS. Column (2): country fixed effects, pooled capacity preservation. Column (3): CP disaggregated into above-the-line (wage subsidies, grants) and below-the-line (loans, guarantees) at face value. Column (4): below-the-line adjusted for 35% guarantee take-up rate (International Monetary Fund, 2022). Column (5): three-way decomposition into above-the-line, government loans (actual disbursements), and credit guarantees (contingent liabilities, adjusted at 35%). Loans are highly significant ($\hat{\kappa}^{\text{loans}} = 0.35$***, $p < 0.001$) while guarantees are insignificant ($p = 0.57$), consistent with the "whatever it takes" logic: guarantees preserve capacity through announcement effects but most are never drawn, generating no realized debt. F^{DI} enters at lag 1. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.5 Identifying Variation in the Debt Equation

Table 14 reports the analogous variance decomposition for the debt equation. Because the debt equation does not include quarter fixed effects—the output gap y_{ik} absorbs common temporal dynamics through the automatic stabilizer channel—the relevant benchmark is within-country, across-quarter variation.

Table 14: Variance Decomposition of Regressors, Debt Equation

	Total SD	Between SD	Within SD	R^2 absorbed (C FE)
Change in debt, Δb_{ik}	2.48	0.54	2.42	0.047
Output gap, y_{ik}	4.77	3.05	3.66	0.401
CP above-the-line, $F_{ik}^{CP,\text{above}}$	0.98	0.27	0.94	0.076
CP loans, $F_{ik}^{CP,\text{loans}}$	1.02	0.30	0.97	0.084
CP guarantees (adj.), $F_{ik}^{CP,\text{guar}}$	0.73	0.18	0.71	0.058
Demand injection, F_{ik}^{DI}	0.71	0.25	0.66	0.127
Health, F_{ik}^{H}	0.38	0.12	0.36	0.104

Notes: Estimation sample (Q1 2020–Q2 2022, $N = 380$). R^2 absorbed reports the share of raw variance explained by country fixed effects alone.

Country fixed effects absorb only a small share of the variance in pandemic-era fiscal variables—between 6% and 13%— because pandemic-specific fiscal instruments had no pre-pandemic counterpart and cross-country level differences are therefore limited. More than 85% of the variance in each fiscal component is identifying.

Table 15 reports residual correlations after country fixed effects. Two features merit discussion. First, the fiscal components are moderately correlated with one another (0.15–0.33), reflecting a common pattern of fiscal generosity: countries that deployed large volumes in one channel also deployed in others. This correlation is a source of multicollinearity that enlarges standard errors on the individual κ coefficients, especially for guarantees, whose residual correlation with above-the-line (0.333) and loans (0.301) is the highest in the matrix. Second, the output gap is negatively correlated with all fiscal components (-0.20 to -0.23), consistent with deeper recessions eliciting larger fiscal responses. The γ_y coefficient is therefore identified from residual variation in y_{ik} net of this common countercyclical deployment pattern.

Table 15: Residual Correlations after Country FE, Debt Equation

	y	above	loans	guar	F^{DI}	F^{H}
y_{ik}	1.000	−0.218	−0.227	−0.215	−0.198	−0.197
$F^{CP,\text{above}}$	−0.218	1.000	0.161	0.333	0.258	0.207
$F^{CP,\text{loans}}$	−0.227	0.161	1.000	0.301	0.219	0.205
$F^{CP,\text{guar}}$	−0.215	0.333	0.301	1.000	0.161	0.152
F^{DI}	−0.198	0.258	0.219	0.161	1.000	0.199
F^{H}	−0.197	0.207	0.205	0.152	0.199	1.000

Notes: Correlations of residuals after country fixed effects demeaning. Estimation sample (Q1 2020–Q2 2022, $N = 380$).

5 Optimization

5.1 Solution Method

With all structural parameters either estimated from the panel (Section 4) or calibrated from the literature (Table 16), the planner's problem (15) is solved numerically using an iterative Linear-Quadratic Regulator (iLQR)—a direct shooting method following Li and Todorov (2004) and Bertsekas (2017). The algorithm alternates between a forward pass that rolls the nonlinear transition system (11)–(14) forward under a candidate control sequence, and a backward pass that solves a time-varying LQR problem around the resulting trajectory to update the controls. Non-negativity constraints on the controls ($S_k, F_k^{CP}, F_k^{DI} \geq 0$) are enforced through an Augmented Lagrangian scheme. The solver is initialized at each country's observed fiscal trajectory, and convergence is verified by solving from alternative initializations.

The planner's weights (w_d, w_y, w_b, W_b) and control penalties (r_S, r_{CP}, r_{DI}) are not estimated but selected to reflect the preferences revealed by observed policy: the weights are calibrated such that the unconstrained aggregate optimum reproduces the OECD-average trajectory of output and debt over Q1 2020–Q2 2022. This revealed-preference calibration serves as the benchmark against which country-specific counterfactuals are evaluated. For the counterfactual exercises, $\{S_k\}$ is fixed at each country's empirically observed trajectory and the induced epidemiological states $\{\theta_k, d_k\}$ enter as predetermined inputs, so that the optimization reduces to the fiscal controls $\{F_k^{CP}, F_k^{DI}\}$ under the same preference weights.

5.2 Calibration

The transition system (11)–(14) contains two sets of structural parameters. The first set—governing epidemiological dynamics and the macroeconomic environment—is calibrated from the existing literature. The second set—governing the transmission of disaggregated fiscal instruments to output and debt—is estimated from the panel data. Table 16 reports the calibrated parameters with sources; Table 17 reports the estimated parameters with standard errors. A complete list of all variables and parameters is provided in Table **??** in the Appendix.

Table 16: Calibrated Parameters[18]

Parameter	Description	Value	Source
ρ_θ	Eff. reproduction rate (quarterly)	1.25–1.70	Dhungel et al. (2022); Liu and Rocklöv (2021)
δ_θ	Infection fatality rate	0.003–0.020	COVID-19 Forecasting Team (2022b); Levin et al. (2020)
ϕ_S	NPI suppression effectiveness	0.4–0.7	Brauner et al. (2021); Haug et al. (2020)
r	Real interest rate (quarterly)	≈ 0.001	OECD (2022)
β	Discount factor (quarterly)	0.99	Standard
c_H	Health expenditure per % infection	0.02	International Monetary Fund (2022); OECD (2023)

Table 17: Estimated Structural Parameters

Parameter	Description	Estimate	Equation
Output transition			
ρ_y	Baseline output persistence	0.311*** (0.085)	(13)
ψ	Lockdown amplification of persistence	0.002* (0.001)	(13)
η_p	Capacity preservation persistence channel	−0.011*** (0.003)	(13)
α_S	Direct output cost of containment	−0.034*** (0.012)	(13)
α_{DI}	Demand injection multiplier (lag 2)	0.185 (0.122)	(13)
β_d	Behavioral response to mortality	−0.021** (0.008)	(13)
Debt transition			
γ_y	Budget semi-elasticity (automatic stabilizer)	0.210*** (0.037)	(14)
κ^{above}	Above-the-line CP cost	0.423* (0.174)	(14)
κ^{loans}	CP loans (actual disbursements)	0.352*** (0.083)	(14)
κ^{guar}	CP guarantees (35% take-up)	0.095 (0.164)	(14)
κ^{DI}	Demand injection fiscal cost	0.181 (0.282)	(14)

Notes: Output parameters from the two-way fixed effects specification (Table 4, column 3). Debt parameters from the three-way CP decomposition (Table 13, column 5). Standard errors clustered at the country level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The estimated parameters identify the two transmission channels that the theoretical framework distinguishes. On the output side, the persistence coefficient $\eta_p = -0.011$ is highly significant, confirming that capacity preservation reduces the effective persistence of the output gap in proportion to the depth of the recession. The demand injection multiplier $\alpha_{DI} = 0.185$ is positive but not statistically distinguishable from zero, consistent with the prediction that demand stimulus is ineffective when supply is physically constrained. On the debt side, the automatic stabilizer $\gamma_y = 0.210$ lies within the OECD range of 0.35–0.65 reported by Price et al. (2015) once scale differences are accounted for, and the CP cost decomposition reveals the expected ordering: above-the-line outlays (0.42) exceed loan disbursements (0.35), which in turn exceed contingent guarantees (0.10, insignificant).

[18]The epidemiological parameters ρ_θ and δ_θ vary across pandemic waves; the ranges reported here span from the second wave (late 2020) through Omicron (Q4 2021). Wave-specific values and their sources are reported in Table **??** in the Appendix.

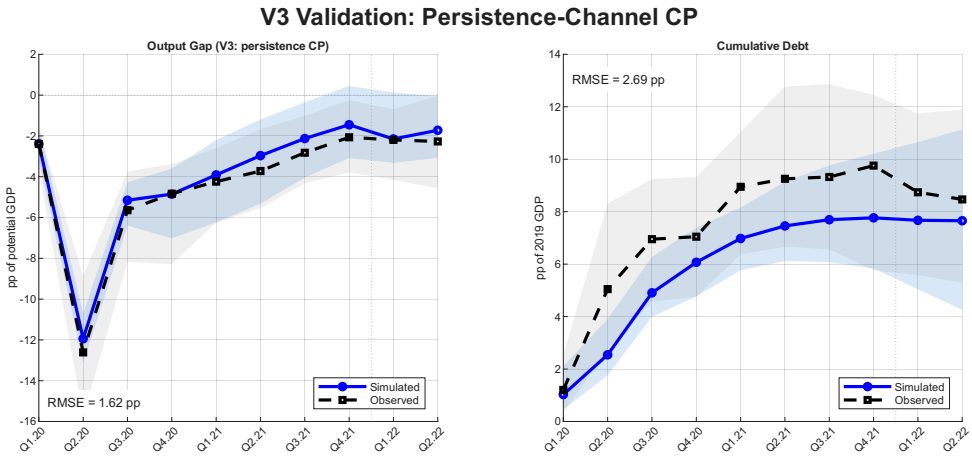

Figure 4: Calibration OECD Average

5.3 Scenarios

5.4 Counterfactuals

6 Conclusion

7 Robustness

Robustness Checks: Summary of Required Tests

1. **Simultaneity of S and F (Referee Q1)**

   - Within-quarter cross-correlation of ΔS and F^{CP} / F^{DI} at weekly frequency (Figure in Appendix **??**)
   - Asymmetric lag structure: F responds to S, not the reverse ($r < 0.05$ at negative lags, $r > 0$ at positive lags)
   - F^{CP} lead test: $F^{CP}_{i,k+1}$ as placebo regressor — expected insignificant

Within-quarter timing. To verify the temporal ordering of containment and fiscal deployment, Figure **??** reports weekly cross-correlations between changes in stringency ΔS_t and fiscal flows F^{CP}_t and F^{DI}_t at lags $h = -6, \ldots, +6$ weeks. The pattern is strongly asymmetric: at negative lags (fiscal deployment preceding containment changes), correlations are near zero or negative ($r < 0.01$ for $h \leq -2$). At lag zero and positive lags (containment leading fiscal deployment), correlations are positive and persistent, peaking contemporaneously at $r = 0.19$ for CP and $r = 0.09$ for DI. Granger causality tests confirm this ordering: ΔS strongly predicts F^{CP} ($F = 40.7$, $p < 0.0001$) and F^{DI} ($F = 6.5$, $p < 0.0001$), while F^{DI} does not predict ΔS ($F = 0.28$, $p = 0.89$). The reverse direction $F^{CP} \to \Delta S$ shows some statistical significance ($F = 6.8$, $p < 0.0001$), consistent with governments announcing lockdown measures and fiscal support packages simultaneously—this does not threaten identification, which relies on the orthogonality of fiscal *composition* to containment intensity, not on the absence of any temporal association between aggregate fiscal deployment and lockdown decisions.

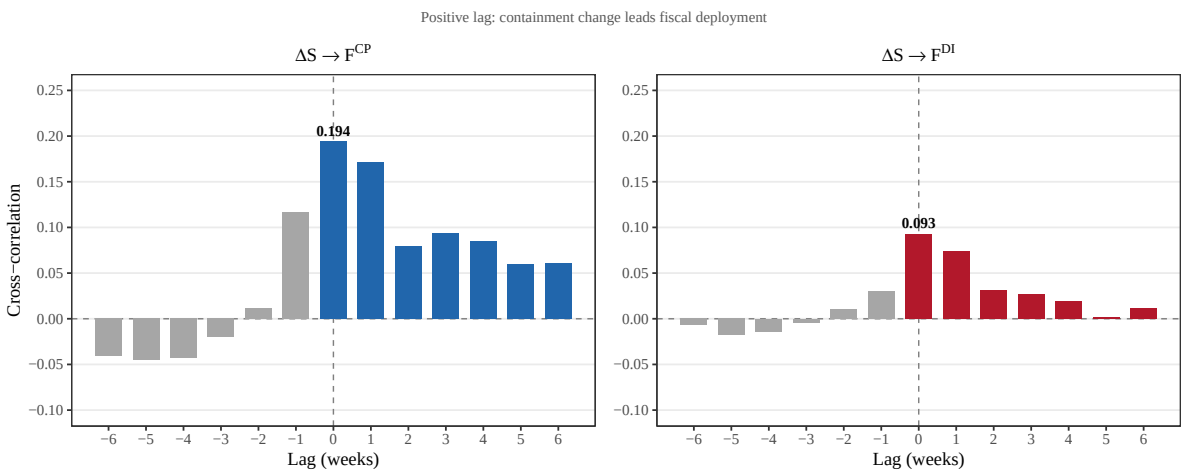

Figure 5: Pandemic Dynamics: Infection Prevalence, Excess Mortality, and Containment

2. **Fear Term Specification (Referee Q3)**

   - Specification without d_{ik}: core coefficients $\hat{\psi}$, $\hat{\tilde{\eta}}$ stronger / unchanged
   - Specification with θ_{ik} instead of d_{ik}
   - Specification with p_{avg} instead of p_{proj}
   - All three: $\hat{\psi}$, $\hat{\tilde{\eta}}$, $\hat{\alpha}_F^{DI}$ qualitatively unchanged
   - Identification argument: $d_{ik} = \delta_\theta \theta_{i,k-1}$ is predetermined relative to $S_{ik} \rightarrow$ not a bad control

3. **Persistence Channel η_p (Referee Q4)**

   - Specification without $F^{CP} \times y_{i,k-1}$: $\hat{\psi}$ and $\hat{\tilde{\eta}}$ stable
   - $\hat{\eta}_p$ significant in baseline, fragile to outlier exclusion and alternative containment measures
   - Interpretation: suggestive evidence consistent with the structural preservation channel; not load-bearing for main results
   - Identification: within-country time-variation in $F^{CP} \times y_{i,k-1}$; GDP publication delay rules out contemporaneous reverse causality from y_{k-1} to F_k^{CP}

4. **Endogeneity of y_{ik} in Debt Equation (Referee Q5)**

   - Reduced-form specification excluding y_{ik}: estimates total debt effect (direct costs + automatic stabilizers combined)
   - $\hat{\kappa}$ in reduced form lower than in baseline $\rightarrow$ confirms self-financing channel
   - Direction of bias: inclusion of y_{ik} overstates direct fiscal cost $\hat{\kappa} \rightarrow$ conservative for the self-financing argument
   - Automatic stabilizer is quasi-mechanical (tax-transfer system), not a behavioral channel; reverse causality ($b \rightarrow y$) implausible at quarterly frequency

5. **Sample Horizon (Referee Q6)**

   - Both equations estimated on $T = 9$ (Q1.2020–Q1.2022, main specification)
   - Both equations estimated on $T = 12$ (Q1.2020–Q4.2022, robustness)
   - Coefficients stable across both horizons

6. **DI Lag Selection (Referee Q7)**

   - All three lag specifications (lag 1, 2, 3) reported in a single table
   - Lag 1: insignificant; Lag 2: significant; Lag 3: insignificant

- AIC/BIC comparison if available; otherwise transparent that selection is based on statistical significance and theoretical priors (**??**)
- DI $\times$ CP complementarity interaction (**?**): $p = 0.022$ — DI effectiveness conditional on CP having preserved supply side

Additional Robustness Checks (from earlier discussion)

[resume]

1. **Standard Error Comparison**
   - HC1 (baseline), HC3, CRV1, CRV3 (Bell–McCaffrey)
   - Wild Cluster Bootstrap (Rademacher, $B = 9{,}999$)
   - Qualitative conclusions unchanged across all SE types
2. **Nickell Bias**
   - $O(1/T) \approx 11\%$; absolute bias < 0.007 at $\hat{\rho}_y \approx 0.06$
   - GMM (Arellano–Bond): $\hat{\psi}$ and $\hat{\hat{\eta}}$ quantitatively unchanged
   - GMM diagnostically sound (AR(2) insignificant, Sargan $p > 0.05$) but CP coefficient inflated $\rightarrow$ weak instruments at $N = 38$
3. **Containment Measure**
   - Population-weighted S (baseline) vs. unweighted national index
   - Google Mobility data in place of S
   - Within-quarter maximum $S_{\max}$ instead of mean S
   - Trade-weighted S (addresses spillover concern)
4. **Output Measure**
   - HP-filtered gap (baseline) vs. YoY GDP growth rate
   - Log gap specification
5. **Debt Measure**
   - Real debt normalized by 2019 GDP (baseline) vs. nominal debt
   - Controlling for inflation rate
6. **Outliers**
   - Excluding Ireland (extreme GDP volatility)

- Excluding Turkey (extreme macro conditions)
- Excluding Estonia (sensitive for κ^{above})

7. **Subsample Splits**
   - High vs. low stringency countries
   - High vs. low social safety net countries
   - High vs. low pre-crisis debt countries
   - Pre-vaccination (Q1.2020–Q4.2020) vs. full sample
8. **Guarantee Take-Up Rate**
   - Baseline: 35% take-up
   - Robustness: 25% and 50% take-up
9. **Fiscal Classification**
   - Alternative classification rules for borderline measures
   - Inclusion of tax deferrals (IMF Cat. 3)
10. **Time Controls in Debt Equation**
    - Linear trend (baseline) vs. quadratic trend
    - Half-year fixed effects
11. **Functional Form**
    - Quadratic F^{CP} term: $p = 0.11$, insignificant $\rightarrow$ linear specification validated
    - Threshold model: insignificant

7.1 Structural Parameters for the iLQR

Table 18 collects the estimated structural parameters that enter the iLQR optimization of Section **??**. The output equation identifies $\hat{\alpha}_S$, $\hat{\psi}$, $\hat{\alpha}_F^{CP}$, $\hat{\eta}$, and $\hat{\alpha}_F^{DI}$. The debt equation identifies $\hat{\gamma}_y$, $\hat{\kappa}_F^{CP}$, and $\hat{\kappa}_F^{DI}$. Health expenditure $\hat{c}_H$ is calibrated from the classified health measures in the database.

These parameters, together with the epidemiological parameters (ρ_θ, ϕ_S, δ_θ) calibrated from the infection and mortality data, fully specify the transition system for the iLQR. The LP impulse responses provide additional dynamic validation: the monotonically accumulating debt IRF ($0.16 \rightarrow 0.63$ over five quarters) and the contemporaneous-then-decaying output IRF confirm that the quarterly transition equations capture the relevant dynamics within the iLQR's planning horizon. The next section takes these estimated parameters to the optimization framework and computes the Pareto frontier across the trilemma dimensions.

Table 18: Estimated Structural Parameters

Parameter	Description	Estimate	Source
$\hat{\alpha}_S$	Direct output cost of containment	-0.028	Table ??, Col. 3
$\hat{\psi}$	Lockdown-induced hysteresis	0.004	Table ??, Col. 3
$\hat{\alpha}_F^{CP}$	CP level effect on output	0.260	Table ??, Col. 3
$\hat{\eta}$	CP×S interaction (cushioning)	-0.006	Table ??, Col. 3
$\hat{\alpha}_F^{DI}$	DI multiplier (lag 2)	0.239	Table ??, Col. 3
$\hat{\gamma}_y$	Automatic stabilizer semi-elasticity	0.207	Table ??, Col. 2
$\hat{\kappa}_F^{CP}$	Fiscal cost of CP (pooled)	0.191	Table ??, Col. 2
$\hat{\kappa}_F^{DI}$	Fiscal cost of DI (lag 1)	0.406	Table ??, Col. 2
$\hat{c}_H$	Health expenditure per unit θ	calibrated	Health measures (Codes 30–34)

Notes: Parameters estimated from the panel specifications in Tables ??–??. $\hat{\kappa}_F^{CP}$ is the pooled estimate; the decomposition into above-the-line (0.464) and below-the-line (0.464) is used in the sensitivity analysis. All estimates use cluster-robust HC1 standard errors.

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8 Appendix

A Structured Comparison

Dimension	**Alvarez et al. (2021)**	**Eichenbaum et al. (2021)**	**Kaplan et al. (2020)**	**Pesenti (2026)**
Epidemiology	Continuous SIR	Discrete SIR	Expanded SIR with occupational contact rates	Reduced-form SIR (ρ_θ, δ_θ, ϕ_S)
Economic model	Linear production	NK–DSGE, representative agent	HANK with income/ wealth inequality	Linear-quadratic state-space
Agent structure	Representative	Representative	Heterogeneous (income, wealth, occupation, sector)	Representative planner
Containment instrument	Lockdown $L_t \in [0,1]$	Pigouvian consumption tax μ_t	Sector-specific lockdown	$S_k \in [0,1]$ with ϕ_S
Fiscal instruments	**None**	**None**	Lump-sum transfers, UI extension	F_k^{DI} (demand injection) + F_k^{CP} (capacity preservation)
Instrument disaggregation	n/a	n/a	Partial (transfers vs. UI)	Full (DI vs. CP by transmission mechanism)
Debt / fiscal sustainability	**No**	**No**	Present but not optimized	**Core objective:** b_k is penalized and endogenous
Automatic stabilizers	No	No	Implicit (progressive taxes in HANK)	Explicit: γ_y creates endogenous fiscal feedback
Output persistence / hysteresis	No (linear, instant recovery)	No (equilibrium recovery)	Partial (wealth effects, scarring through inequality)	Explicit: $\rho_y^{\text{eff}}(S_k, F_k^{CP})$

Dimension	Alvarez	Eichenbaum	Kaplan	Pesenti
Optimization method	Pontryagin maximum principle (continuous)	Competitive equilibrium + Ramsey problem	Calibrated scenarios (no joint optimization)	iterative LQR (discrete-time, Riccati)
State vector	(S, I, R, D)	(S, I, R, D) + macro	(S, I, R, D) + full HANK distribution	$(y_k, d_k, b_k, \theta_k)$
Control vector	L_t (scalar)	μ_t (scalar)	Scenarios (not optimized)	$(S_k, F_k^{DI}, F_k^{CP})$ (3-dim)
Objective function	$\min \int [wL + \theta \dot{D}]\, dt$ (2 terms)	Social welfare (consumption utility)	Distributional welfare	$\min \sum \beta^k [w_y y_k^2 + w_d d_k^2 + w_b b_k^2 + R\|u_k\|^2]$ (3 state penalties)
Trade-offs captured	Health ↔ Economy (2D)	Health ↔ Economy (2D)	Health ↔ Economy ↔ Distribution (2.5D)	Health ↔ Economy ↔ Fiscal (3D trilemma)
Empirical calibration	U.S. parameters, COVID-19	U.S. parameters, COVID-19	U.S. calibration, HANK	38 OECD countries, LP estimation, micro database
Cross-country	No	No	No	Yes (38 OECD)
Counterfactual exercises	Optimal vs. no lockdown	CE vs. optimal containment	Scenario comparison	Optimal composition vs. observed; joint vs. conditional optimum

B External Validation: The Trilemma in Comparative Perspective

To establish that the pandemic trilemma represents a structurally unique configuration rather than a generic feature of crisis management, I apply the four constitutive properties as a diagnostic framework to four alternative crisis types: armed conflict, natural disasters,

financial crises, and climate change. The objective is to determine whether the specific combination of properties—and their circular interdependence—arises in other contexts. Table 21 summarizes the results.

Armed Conflict

Armed conflict shares certain surface-level features with the pandemic: it can involve escalatory dynamics and generates fiscal pressures through military expenditure, activating both the debt accumulation channel (Property 3) and automatic stabilizers (Property 4). However, Property 1 is only partially satisfied. While conflict can escalate, the threat does not exhibit the autonomous, biological growth dynamic of $\rho_\theta > 1$: military escalation is mediated by strategic decisions, not by an epidemiological reproduction rate that compounds independently of human choice.

The critical distinction lies in Property 2. Military mobilization and defense spending are typically *expansionary*: wartime production absorbs idle resources, stimulates aggregate demand, and increases employment. The instrument that suppresses the threat operates in the *same* direction as economic stabilization rather than against it—violating the structural requirement that $\alpha_S > 0$ and $\phi_S > 0$ operate through the same policy variable. Furthermore, the nature of the shock differs fundamentally: war destroys physical capital (K), whereas a pandemic paralyzes economic flows while leaving the capital stock intact. The reconstruction problem following a war—rebuilding K toward its pre-war level—is qualitatively different from the restoration problem following a pandemic, where the objective is to return the flow variables (y_k, θ_k) to their pre-crisis steady state. Because the suppression instrument is not inherently contractionary, the circular linkage that generates the trilemma does not arise.

Natural Disasters

Natural disasters constitute the purest form of exogenous shock, but they violate Property 1 in a decisive respect: they are *impulse* shocks with $\rho < 1$. An earthquake or flood strikes, causes damage, and ceases. The shock does not grow endogenously; it dissipates without intervention. Consequently, there is no necessity for a suppression instrument: the planner faces a reconstruction problem, not a containment problem. Property 2 is therefore vacuous—since no active suppression is required, the question of whether suppression is contractionary does not arise. While the fiscal pressures from reconstruction can be substantial (Properties 3 and 4 are present), the absence of the self-reinforcing dynamic ($\rho_\theta > 1$) and the contractionary suppression instrument means that the circular linkage cannot form. The planner faces a costly rebuilding challenge, but not a trilemma.

Financial Crises

Financial crises violate Property 1 at its foundation: the shock is *endogenous* to the economic system. It originates within the financial sector through excessive leverage, asset price misalignments, or institutional failures. Even the international propagation that characterized the 2008 crisis—transmission through interbank lending channels and cross-border contagion—represents the spread of endogenous instabilities through system-internal linkages rather than the arrival of an external shock (ε_k).

Property 2 is violated in a structural sense. The primary intervention instruments—bank recapitalization, liquidity provision, and asset purchases—are *expansionary* for the real economy, or at minimum prevent a contractionary collapse. This reverses the causal logic of the pandemic trilemma: in a pandemic, the suppression of the shock *itself* is contractionary ($\alpha_S > 0$), whereas in financial crises, contractionary outcomes arise from the *failure* to intervene or from subsequent fiscal consolidation. Austerity policies following the European sovereign debt crisis were contractionary, but they constituted a response to the fiscal legacy of the crisis, not the crisis-fighting instrument per se. The trilemma structure does not emerge because the instrument that addresses the threat works *with* rather than *against* economic stabilization.

Climate Change

Climate change is the most structurally analogous case and merits extended discussion.

Property 1 is satisfied: greenhouse gas accumulation exhibits self-reinforcing dynamics analogous to $\rho_\theta > 1$. CO_2 concentrations compound in the atmosphere, positive feedback loops (permafrost thawing, ice-albedo effects) accelerate warming, and potential tipping points imply that the process may become irreversible beyond certain thresholds. The lethality channel ($\delta_\theta > 0$) operates through climate-related mortality: heat stress, extreme weather events, and food system disruptions.

Property 2 is satisfied: the suppression instruments—carbon taxation, emission standards, and the phase-out of fossil fuel infrastructure—are contractionary for carbon-intensive sectors and, at least transitionally, for aggregate output. As in the pandemic, the mechanism that suppresses the threat simultaneously restricts economic activity ($\alpha_S > 0$).

Property 3 is satisfied: the transition to a low-carbon economy requires substantial public investment in infrastructure, technology, and social compensation programs, generating intertemporal fiscal liabilities ($\kappa_F > 0$).

Property 4 is satisfied: an economic slowdown induced by climate policy activates the same automatic stabilizer channel ($-\gamma_y\, y_k$) as a pandemic-induced contraction.

All four constitutive properties are present, and a circular linkage analogous to the pandemic trilemma can be identified: emission reduction contracts the economy, requiring fiscal compensation, which generates debt, which constrains future fiscal space for the

transition. However, three qualitative differences distinguish the climate case and preclude a direct application of the framework developed here:

1. **Time scale.** The pandemic trilemma operates on a horizon of quarters to years; the climate trilemma on decades. This transforms the discount factor β from a technical parameter into a normative choice about intergenerational equity, fundamentally altering the political economy of the optimization problem.

2. **Observability and attribution.** Pandemic mortality is immediate, individually attributable, and politically salient—the lag from S_k to d_{k+2} spans weeks. Climate-related mortality is diffuse, statistically distributed across populations, and temporally remote. The policy lag between suppression and observable health outcomes is orders of magnitude longer, rendering the political economy of sustained suppression qualitatively more difficult.

3. **Terminal condition.** A pandemic admits a finite planning horizon: vaccination exogenously decouples mortality from infection (reducing δ_θ and ρ_θ), dissolving the trilemma at the terminal date and leaving only the fiscal legacy. Climate change admits no analogous resolution; the suppression effort must be sustained indefinitely. The terminal condition that resolves the pandemic trilemma—the exogenous dissolution of the epidemiological constraint through vaccination, as discussed in Property 2—has no climate analogue.

These differences do not invalidate the structural analogy but imply that a climate trilemma framework would require fundamentally different modeling choices regarding the planning horizon, the discount structure, and the terminal condition. The pandemic setting therefore offers a more tractable environment in which to develop and test the trilemma framework, with potential for future extension to the climate domain.

Table 20: Diagnostic Comparison of Crisis Types

	Property 1 Exogenous, $\rho > 1$	Property 2 Contractionary	Property 3 Debt-generating	Property 4 Auto. Stabilizers
Pandemic	✓	✓	✓	✓
Armed Conflict	Partial	×	✓	✓
Natural Disaster	×	Vacuous	✓	✓
Financial Crisis	×	×	✓	✓
Climate Change	✓	✓	✓	✓

B.1 Comparative Summary

Table 21 synthesises the diagnostic assessment across crisis types.

Table 21: Constitutive Properties of the Trilemma Across Crisis Types

Property	**Pandemic**	**War**	**Nat. Dis.**	**Financial**	**Climate**
1. Self-reinforcing exogenous shock	✓	∼	×	×	✓
2. Contractionary suppression instr.	✓	×	n/a	×	✓
3. Intertemporal fiscal liability	✓	✓	✓	✓	✓
4. Endogenous fiscal feedback	✓	∼	∼	✓	✓
5. Irreducible policy lag	✓	×	×	∼	✓
Trilemma structure	**Yes**	**No**	**No**	**No**	**Potentially**

Notes: ✓ = property satisfied; × = property not satisfied; ∼ = partially satisfied; n/a = not applicable. The trilemma requires the simultaneous presence of all five properties *and* their circular interdependence.

The analysis confirms that the pandemic trilemma is structurally unique among the crisis types considered. Climate change is the only context in which an analogous trilemma structure may emerge, although the fundamental differences in time scale, observability, and reversibility imply that the climate trilemma—should it be formalised—would require substantial modifications to the analytical framework.

C Derivation of Policy Necessity and Economic Impact

Due to the transmission lag, immediate mortality is predetermined. The planner controls future mortality d_{k+2} by suppressing current viral transmission. Based on the transmission dynamics defined in Eq. (**??**), the causal chain is:

$$d_{k+2} = \delta_\theta \cdot \underbrace{\rho_\theta(1 - \phi_S S_k)\theta_k}_{\theta_{k+1}} \tag{18}$$

Consequently, the necessity of the policy instrument S_k depends on the conjunction of the lethality parameter δ_θ and the viral growth potential ρ_θ.

Let economic output Y_k be a decreasing function of suppression S_k. The policy constraint on output is strictly binding if and only if the virus is lethal ($\delta_\theta > 0$) and transmissible ($\rho_\theta > 0$).

Proof. We analyze the marginal effect of the suppression policy S_k on the health outcome

d_{k+2}. Differentiating Eq. (18) with respect to S_k yields:

$$\frac{\partial d_{k+2}}{\partial S_k} = \delta_\theta \cdot \rho_\theta \cdot (-\phi_S) \cdot \theta_k = -\delta_\theta \rho_\theta \phi_S \theta_k \tag{19}$$

The marginal health benefit of suppression (reducing deaths) is non-zero if and only if the derivative is non-zero.

- **Case $\delta_\theta = 0$ (Non-lethal):** The derivative becomes zero. Suppression saves no lives. The optimal strategy is $S_k = 0$ to maximize output.
- **Case $\delta_\theta > 0$ (Lethal):** Given an active epidemic ($\theta_k > 0$) and effective NPIs ($\phi_S > 0$), the derivative is strictly negative. Increasing S_k significantly reduces future mortality.

Therefore, the planner is forced to accept economic costs (set $S_k > 0$) solely because $\delta_\theta > 0$. The trade-off is conditional on the simultaneous existence of lethality and transmissibility. □

D Analytical Framework

D.1 Solution Method

The planner chooses three instruments simultaneously at each quarter over a finite horizon, where the effectiveness of each instrument depends on the current state of the crisis and transmits to outcomes with heterogeneous time lags. The three objective dimensions—health, output, and debt—cannot be optimized independently, as every instrument affects multiple dimensions simultaneously. Exogenous pandemic waves perturb the system stochastically, and all instruments are subject to physical bounds. Standard methods cannot solve this problem: LQR requires linear dynamics, Pontryagin's maximum principle requires analytical costate solutions unavailable for four bilinear terms, and value function iteration is infeasible over a continuous five-dimensional state space. Therefore I solve the problem using the iterative Linear-Quadratic Regulator (iLQR) of Li and Todorov (2004), a trajectory optimization method from the nonlinear control literature that handles precisely this problem class.[19]

The algorithm proceeds in three steps. First, starting from an initial control sequence $\{\bar{\boldsymbol{u}}_k^{(j)}\}$, the nonlinear dynamics are rolled forward to generate a nominal trajectory $\{\bar{\boldsymbol{x}}_k^{(j)}\}$.[20] Second, the dynamics are linearized around this trajectory—computing the

[19]The notation follows Bertsekas (2017, Chapters 3–4). Abhijeet and Chakravorty (2025) show that iLQR is equivalent to Sequential Quadratic Programming applied to the optimal control problem, with guaranteed cost descent at each iteration.

[20]The bilinear terms render the full problem non-convex. The iterative linearization converges to a local optimum; global optimality is verified by solving from multiple initialization trajectories (Appendix E.6).

Jacobians $\boldsymbol{A}_k = \partial \boldsymbol{f}/\partial \boldsymbol{x}\big|_{\bar{\boldsymbol{x}}_k,\bar{\boldsymbol{u}}_k}$ and $\boldsymbol{B}_k = \partial \boldsymbol{f}/\partial \boldsymbol{u}\big|_{\bar{\boldsymbol{x}}_k,\bar{\boldsymbol{u}}_k}$—and a time-varying LQR backward pass computes feedback gains $\boldsymbol{K}_k$ and feedforward corrections $\boldsymbol{k}_k$ via the Riccati recursion. Third, the updated control sequence

$$\boldsymbol{u}_k^{(j+1)} = \bar{\boldsymbol{u}}_k^{(j)} + \alpha\, \boldsymbol{k}_k + \boldsymbol{K}_k\big(\boldsymbol{x}_k^{(j+1)} - \bar{\boldsymbol{x}}_k^{(j)}\big) \tag{20}$$

is applied to the full nonlinear dynamics with a line search parameter $\alpha \in (0, 1]$, generating a new trajectory. Box constraints on the controls—$S_k \in [0, 1]$, $F_k^{DI} \geq 0$, $F_k^{CP} \geq 0$—are enforced through an Augmented Lagrangian scheme (Howell et al., 2019) with outer multiplier updates. Under certainty equivalence (Bertsekas, 2017, Chapter 3.1), stochastic shocks do not alter the optimal policy rule: the planner optimizes against the expected trajectory, and the feedback gains $\boldsymbol{K}_k$ provide automatic adjustment to realized deviations. The complete derivation—including explicit Jacobians, Q-function expansion, regularization, and convergence criteria—is provided in Appendix **??**. The model is now fully specified: the transition system, objective function, and solution method are determined up to the structural parameters that govern fiscal transmission and debt dynamics. Section 4 identifies these parameters.

D.2 Transition System and Augmented State Space

The transition system from Section **??** contains the lagged term F_{k-1}^{DI} in the output equation, which violates the first-order Markov property required by the iLQR algorithm. Introducing the auxiliary state $z_k \equiv F_{k-1}^{DI}$ and the augmented state vector $\tilde{\boldsymbol{x}}_k = (y_k,\, d_k,\, b_k,\, \theta_k,\, z_k)' \in \mathbb{R}^5$, the system becomes:

$$y_{k+1} = \rho_y\, y_k + (\psi\, y_k - \alpha_S)\, S_k + (\alpha_F^{CP} + \tilde{\eta}\, S_k - \eta_p\, y_k)\, F_k^{CP} + \alpha_F^{DI}\, z_k + \beta_d\, d_k \tag{21}$$

$$d_{k+1} = \delta_\theta(k)\, \theta_k \tag{22}$$

$$b_{k+1} = (1 + r)\, b_k - \gamma_y\, y_k + \kappa^{CP}\, F_k^{CP} + \kappa^{DI}\, F_k^{DI} + c_H\, \theta_k \tag{23}$$

$$\theta_{k+1} = \rho_\theta(k)\big(1 - \phi_S\, S_k\big)\, \theta_k + \varepsilon_k \tag{24}$$

$$z_{k+1} = F_k^{DI} \tag{25}$$

The disturbance selection vector is $\tilde{\boldsymbol{G}} = (0,\, 0,\, 0,\, 1,\, 0)' \in \mathbb{R}^5$, so that $\tilde{\boldsymbol{x}}_{k+1} = f(\tilde{\boldsymbol{x}}_k,\, \boldsymbol{u}_k,\, k) + \tilde{\boldsymbol{G}}\, \varepsilon_k$. The epidemiological parameters $\rho_\theta(k)$ and $\delta_\theta(k)$ are time-varying across pandemic waves; their wave-specific values are reported in Table 24.

The asymmetry in the timing of fiscal instruments—$\kappa^{DI}\, F_k^{DI}$ (contemporaneous) in the debt equation versus $\alpha_F^{DI}\, z_k = \alpha_F^{DI}\, F_{k-1}^{DI}$ (lagged) in the output equation—reflects the institutional sequence: demand injection commits the budget at authorization (k), but reaches aggregate spending with a disbursement-to-expenditure delay identified empiri-

cally as approximately two quarters (Section 4).

D.3 Jacobian Matrices

The linearization around a nominal trajectory $(\bar{\boldsymbol{x}}_k, \bar{\boldsymbol{u}}_k)$ yields the time-varying affine approximation $\delta\boldsymbol{x}_{k+1} \approx \tilde{\boldsymbol{A}}_k\,\delta\boldsymbol{x}_k + \tilde{\boldsymbol{B}}_k\,\delta\boldsymbol{u}_k$, where:

$$\tilde{\boldsymbol{A}}_k = \left.\frac{\partial f}{\partial \boldsymbol{x}_k}\right|_{(\bar{\boldsymbol{x}}_k,\,\bar{\boldsymbol{u}}_k)} = \begin{pmatrix} \rho_y + \psi\,\bar{S}_k - \eta_p\,\bar{F}_k^{CP} & \beta_d & 0 & 0 & \alpha_F^{DI} \\ 0 & 0 & 0 & \delta_\theta(k) & 0 \\ -\gamma_y & 0 & (1+r) & c_H & 0 \\ 0 & 0 & 0 & \rho_\theta(k)(1-\phi_S\,\bar{S}_k) & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \tag{26}$$

$$\tilde{\boldsymbol{B}}_k = \left.\frac{\partial f}{\partial \boldsymbol{u}_k}\right|_{(\bar{\boldsymbol{x}}_k,\,\bar{\boldsymbol{u}}_k)} = \begin{pmatrix} \psi\,\bar{y}_k - \alpha_S + \tilde{\eta}\,\bar{F}_k^{CP} & 0 & \alpha_F^{CP} + \tilde{\eta}\,\bar{S}_k - \eta_p\,\bar{y}_k \\ 0 & 0 & 0 \\ 0 & \kappa^{DI} & \kappa^{CP} \\ -\rho_\theta(k)\,\phi_S\,\bar{\theta}_k & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \tag{27}$$

The output row of $\tilde{\boldsymbol{B}}_k$ encodes the three CP channels established in Property 3: $\partial y_{k+1}/\partial F_k^{CP} = \alpha_F^{CP} + \tilde{\eta}\,\bar{S}_k - \eta_p\,\bar{y}_k$, where α_F^{CP} is the level effect (Section **??**), $\tilde{\eta}\,\bar{S}_k$ is the contemporaneous cushioning channel (Section **??**), and $-\eta_p\,\bar{y}_k$ is the structural persistence reduction (Section **??**). The containment column $\tilde{\boldsymbol{B}}_k[\cdot, 1]$ similarly contains the cushioning interaction $\tilde{\eta}\,\bar{F}_k^{CP}$: the marginal output cost of lockdown is reduced when CP is deployed, formalizing the enabling role of capacity preservation for containment policy.

D.4 Eigenvalue Structure and Stability

The augmented system matrix $\tilde{\boldsymbol{A}}_k$ is block-triangular, so its eigenvalues coincide with the diagonal entries:

$$\lambda_1^k = \rho_y + \psi\,\bar{S}_k - \eta_p\,\bar{F}_k^{CP} \qquad \text{(output persistence)} \tag{28}$$

$$\lambda_2 = 0 \qquad \text{(mortality: memoryless)} \tag{29}$$

$$\lambda_3 = 1 + r \qquad \text{(debt: unconditionally unstable)} \tag{30}$$

$$\lambda_4^k = \rho_\theta(k)\big(1 - \phi_S\,\bar{S}_k\big) \qquad \text{(epidemiology: conditionally unstable)} \tag{31}$$

$$\lambda_5 = 0 \qquad \text{(auxiliary state: bookkeeping)} \tag{32}$$

Unconditionally unstable: debt. For any positive real interest rate, $\lambda_3 = 1 + r > 1$ lies outside the unit circle regardless of the planner's policy choices. No element of the control vector enters $\tilde{\boldsymbol{A}}_k[3,3]$. During the pandemic, both off-diagonal forces—the automatic stabilizer channel ($-\gamma_y\, y_k > 0$ when $y_k < 0$) and discretionary fiscal support ($\kappa^{DI}\, F_k^{DI} + \kappa^{CP}\, F_k^{CP} > 0$)—reinforce the explosive tendency.

Conditionally unstable: epidemiology. The infection eigenvalue exceeds unity when suppression is insufficient:

$$\lambda_4^k \geq 1 \quad \Longleftrightarrow \quad S_k < \underline{S} \equiv \frac{1 - 1/\rho_\theta(k)}{\phi_S} \tag{33}$$

Below this threshold the epidemic grows despite intervention. The threshold is time-variant across waves (through $\rho_\theta(k)$) but independent of the current prevalence θ_k: the *level* of infections determines the death toll, while the *stability* of the epidemiological state depends only on the structural parameters ρ_θ and ϕ_S.

Conditionally unstable: output. In the absence of a pandemic ($S_k = 0$, $F_k^{CP} = 0$), $\lambda_1 = \rho_y < 1$ and the economy is self-stabilizing. Containment increases the eigenvalue by $\psi\, S_k$ (hysteresis); capacity preservation reduces it by $\eta_p\, F_k^{CP}$ (persistence reduction). The critical condition is:

$$\lambda_1^k \geq 1 \quad \Longleftrightarrow \quad F_k^{CP} < \underline{F}^{CP}(S_k) \equiv \frac{\psi\, S_k - (1 - \rho_y)}{\eta_p} \tag{34}$$

This threshold is positive only when $\psi\, S_k > 1 - \rho_y$, i.e., when the lockdown overcomes the economy's natural stability margin. A stricter lockdown requires more capacity preservation to prevent the output state from becoming explosive.

Cascading instabilities. The planner faces a cascading stabilization problem:

1. The epidemiological instability ($\lambda_4 > 1$) necessitates containment ($S_k > \underline{S}$).
2. Containment destabilizes the output dimension (λ_1 increases toward and potentially beyond unity), necessitating capacity preservation ($F_k^{CP} > \underline{F}^{CP}(S_k)$).
3. The resulting output gap ($y_k < 0$) widens the deficit through automatic stabilizers ($-\gamma_y\, y_k > 0$), necessitating demand injection ($F_k^{DI} > 0$) to close the gap.
4. Both fiscal instruments augment the unconditionally unstable debt state, which cannot be stabilized by any available control.

Each stabilizing action on one dimension contributes to the instability of at least one other. This cascade formalizes the trilemma as a dynamical systems problem.

D.5 Controllability

Since $\text{rank}(\tilde{\boldsymbol{B}}_k) = 3 < 5 = \dim(\tilde{\boldsymbol{x}})$, the mortality state d_k and the auxiliary state z_k are not directly controllable. However, the controllability matrix $\mathcal{C}_k = [\tilde{\boldsymbol{B}}_k \;\; \tilde{\boldsymbol{A}}_k\tilde{\boldsymbol{B}}_k \;\; \cdots \;\; \tilde{\boldsymbol{A}}_k^4\tilde{\boldsymbol{B}}_k]$ has full row rank under non-degeneracy conditions (δ_θ, ϕ_S, α_S, α_F^{DI}, κ^{DI}, $\kappa^{CP} \neq 0$ and $\bar{\theta}_k$, $\bar{y}_k \neq 0$). Mortality is reachable through $\tilde{\boldsymbol{A}}_k\tilde{\boldsymbol{B}}_k$ via the composite parameter $\delta_\theta\,\rho_\theta\,\phi_S\,\bar{\theta}_k$, confirming the two-period transmission lag $S_k \to \theta_{k+1} \to d_{k+2}$.

D.6 Optimality Conditions

The planner minimizes the intertemporal cost (15) subject to the augmented transition (21)–(25). The Bellman equation for the cost-to-go $V_k(\boldsymbol{x}_k)$ is:

$$V_k(\boldsymbol{x}_k) = \min_{\boldsymbol{u}_k} \left\{ \tfrac{1}{2}\boldsymbol{x}_k'\,\boldsymbol{Q}_k\,\boldsymbol{x}_k + \tfrac{1}{2}\boldsymbol{u}_k'\,\boldsymbol{R}_k\,\boldsymbol{u}_k + \mathbb{E}\big[V_{k+1}\big(f(\boldsymbol{x}_k,\,\boldsymbol{u}_k) + \tilde{\boldsymbol{G}}\,\varepsilon_k\big)\big] \right\} \tag{35}$$

with terminal condition $V_N(\boldsymbol{x}_N) = \frac{1}{2}\boldsymbol{x}_N'\,\boldsymbol{Q}_N\,\boldsymbol{x}_N$ and weight matrices as in (**??**). The first-order condition yields:

$$\boldsymbol{R}_k\,\boldsymbol{u}_k + \tilde{\boldsymbol{B}}_k'\,\nabla_{\boldsymbol{x}} V_{k+1}\big|_{\boldsymbol{x}_{k+1}=f(\boldsymbol{x}_k,\,\boldsymbol{u}_k)} = 0 \tag{36}$$

Defining the costate vector $\boldsymbol{p}_{k+1} = \nabla_{\boldsymbol{x}} V_{k+1}(\boldsymbol{x}_{k+1})$ with components $(p_{k+1}^y,\, p_{k+1}^d,\, p_{k+1}^b,\, p_{k+1}^\theta,\, p_{k+1}^z)'$ and expanding (36) using the structure of $\tilde{\boldsymbol{B}}_k$:

Optimal lockdown intensity:

$$S_k^* = -\frac{1}{2\,r_S}\left[\big(\psi\,y_k - \alpha_S + \tilde{\eta}\,F_k^{CP}\big)\,p_{k+1}^y \;-\; \rho_\theta\,\phi_S\,\theta_k\,p_{k+1}^\theta\right] \tag{37}$$

The lockdown balances the output cost against the epidemiological benefit. The term $\tilde{\eta}\,F_k^{CP}$ in the output channel reflects the enabling role of capacity preservation: at higher CP deployment, the marginal output cost of containment is reduced (since $\tilde{\eta} < 0$), permitting stricter lockdowns at lower welfare cost.

Optimal demand injection:

$$F_k^{DI*} = -\frac{1}{2\,r_{DI}}\left[\kappa^{DI}\,p_{k+1}^b \;+\; p_{k+1}^z\right] \tag{38}$$

In the augmented system, DI does not affect output contemporaneously but operates through the auxiliary state $z_{k+1} = F_k^{DI}$. The shadow price p_{k+1}^z captures the discounted future output benefit of current DI (transmitted via $\alpha_F^{DI}\,z_{k+1}$ in the next period's output equation). The debt cost $\kappa^{DI}\,p_{k+1}^b$ provides the counterweight.

Optimal capacity preservation:

$$F_k^{CP*} = -\frac{1}{2\,r_{CP}}\left[\left(\alpha_F^{CP} + \tilde{\eta}\,S_k - \eta_p\,y_k\right)p_{k+1}^y \;+\; \kappa^{CP}\,p_{k+1}^b\right] \tag{39}$$

The output benefit of CP operates through all three channels simultaneously. The level effect $\alpha_F^{CP}\,p_{k+1}^y$ is state-independent and provides a baseline rationale for CP deployment in any recession. The cushioning channel $\tilde{\eta}\,S_k\,p_{k+1}^y$ is activated by active containment—CP is more valuable when lockdowns are in force. The persistence channel $-\eta_p\,y_k\,p_{k+1}^y$ scales with the depth of the recession: in a deeper downturn ($y_k \ll 0$), $-\eta_p\,y_k > 0$ is larger, increasing the marginal value of CP. The debt cost $\kappa^{CP}\,p_{k+1}^b$ restrains deployment.

Optimal fiscal composition. From (38) and (39), the relative deployment at the optimum satisfies:

$$\frac{F_k^{DI*}}{F_k^{CP*}} = \frac{r_{CP}}{r_{DI}} \cdot \frac{\kappa^{DI}\,p_{k+1}^b + p_{k+1}^z}{\left(\alpha_F^{CP} + \tilde{\eta}\,S_k - \eta_p\,y_k\right)p_{k+1}^y + \kappa^{CP}\,p_{k+1}^b} \tag{40}$$

Since the CP output benefit $(\alpha_F^{CP} + \tilde{\eta}\,S_k - \eta_p\,y_k)$ is increasing in $|y_k|$ (via the $-\eta_p\,y_k$ term when $y_k < 0$) while the DI benefit p_{k+1}^z is approximately constant in y_k, the ratio is decreasing in the depth of the recession: the optimal fiscal mix shifts toward capacity preservation as the crisis deepens. This confirms the result derived from the transition structure in Section 2: CP becomes relatively more valuable under deep recessions with active containment, while DI provides constant marginal returns regardless of the state.

Complementarity. The optimal lockdown (37) depends on p_{k+1}^y and p_{k+1}^θ, which are themselves functions of the entire future trajectory—including future fiscal policy. Through the costate recursion, higher expected future CP reduces λ_1^k toward the stable region (via η_p), lowering the cost-to-go associated with a given output gap and thereby reducing $|p_{k+1}^y|$. This permits stricter lockdowns at lower welfare cost, formalizing the enabling role of CP for containment policy.

D.7 Three-Period Forward Iteration

This section iterates the augmented transition system (21)–(25) forward for three periods from the pre-pandemic steady state. The exercise makes the temporal transmission of each instrument visible and identifies the period at which each channel first contributes to each state variable. To isolate the mechanism, I suppress the health expenditure term $c_H\,\theta_k$ (it enters additively and does not interact with any control), the fear channel $\beta_d\,d_k$ (it enters as a predetermined state-to-state coupling), and set $\varepsilon_k = 0$.

Table D.7 summarizes the marginal effects before the algebraic derivation.

	$k+1$	$k+2$	$k+3$
Containment S_k			
Output	$\psi\, y_k - \alpha_S$	propagated $\times\, \rho^{\text{eff}}$	propagated $\times\, (\rho^{\text{eff}})^2$
Debt	via $-\gamma_y\, y_{k+1}$	via $-\gamma_y\, y_{k+2}$	cumulative
Capacity Preservation F_k^{CP}			
Output	$\alpha_F^{CP} + \tilde{\eta}\, S_k - \eta_p\, y_k$	propagated $\times\, \rho^{\text{eff}}$	propagated $\times\, (\rho^{\text{eff}})^2$
Debt	$+\kappa^{CP}$	$(1{+}r)\kappa^{CP} - \gamma_y$ [output effect]	cumulative
Demand Injection F_k^{DI}			
Output	0 (stored as z_{k+1})	$+\alpha_F^{DI}$	propagated $\times\, \rho^{\text{eff}}$
Debt	$+\kappa^{DI}$	$(1{+}r)\,\kappa^{DI}$	$(1{+}r)^2\kappa^{DI} - \gamma_y\, \alpha_F^{DI}$

I initialize at the pre-pandemic steady state $\tilde{\boldsymbol{x}}_0 = (0,\, 0,\, 0,\, \theta_0,\, 0)'$, where $\theta_0 > 0$ represents the initial outbreak and $z_0 = 0$ reflects the absence of prior fiscal deployment.

Period 0 → 1

From $y_0 = 0$, $b_0 = 0$, $z_0 = 0$:

$$y_1 = -\alpha_S\, S_0 \;+\; \big(\alpha_F^{CP} + \tilde{\eta}\, S_0\big)\, F_0^{CP} \tag{41}$$

$$d_1 = \delta_\theta\, \theta_0 \tag{42}$$

$$b_1 = \kappa^{DI}\, F_0^{DI} \;+\; \kappa^{CP}\, F_0^{CP} \tag{43}$$

$$\theta_1 = \rho_\theta\big(1 - \phi_S\, S_0\big)\, \theta_0 \tag{44}$$

$$z_1 = F_0^{DI} \tag{45}$$

Interpretation. The hysteresis channel ($\psi\, S_0\, y_0 = 0$) and the persistence-reduction channel ($\eta_p\, y_0\, F_0^{CP} = 0$) are both inoperative when the economy starts at potential—they activate only when an output gap exists. However, unlike the single-channel specification, the *level effect* α_F^{CP} and the *cushioning channel* $\tilde{\eta}\, S_0$ are operative from the first period. These channels transmit contemporaneously: CP deployed alongside a lockdown partially offsets the contractionary impact within the same quarter. This is the empirical content of Property 3: at the estimated $\tilde{\eta} < 0$, the cushioning channel *reduces* the net CP effect under strict containment (the physical constraint on economic activity limits how much CP can achieve), while the level effect provides a baseline output benefit regardless of the containment regime.

Demand injection has no output effect in period 1 ($\alpha_F^{DI}\, z_0 = 0$) because no prior DI exists. Its output effect materializes with a two-period lag: F_0^{DI} is stored as z_1 and enters

y_2 in the next transition. DI does, however, commit the budget immediately ($\kappa^{DI} F_0^{DI}$ in (43)), reflecting the institutional reality that fiscal authorization precedes disbursement.

Mortality in period 1 is predetermined: $d_1 = \delta_\theta \theta_0$ depends only on the initial infection level and is independent of all controls.

Period 1 → 2

Substituting (41)–(45) into the transition system and defining $\rho_1^{\text{eff}} \equiv \rho_y + \psi S_1 - \eta_p F_1^{CP}$:

$$\begin{aligned} y_2 &= \rho_1^{\text{eff}} y_1 \;-\; \alpha_S S_1 \;+\; \big(\alpha_F^{CP} + \tilde{\eta} S_1\big) F_1^{CP} \;+\; \alpha_F^{DI} F_0^{DI} \\ &= \rho_1^{\text{eff}} \big[-\alpha_S S_0 + \big(\alpha_F^{CP} + \tilde{\eta} S_0\big) F_0^{CP}\big] \;-\; \alpha_S S_1 \;+\; \big(\alpha_F^{CP} + \tilde{\eta} S_1\big) F_1^{CP} \;+\; \alpha_F^{DI} F_0^{DI} \end{aligned} \tag{46}$$

$$d_2 = \delta_\theta \theta_1 = \delta_\theta \rho_\theta \big(1 - \phi_S S_0\big) \theta_0 \tag{47}$$

$$\begin{aligned} b_2 &= (1+r) b_1 \;-\; \gamma_y y_1 \;+\; \kappa^{DI} F_1^{DI} \;+\; \kappa^{CP} F_1^{CP} \\ &= (1+r)\big[\kappa^{DI} F_0^{DI} + \kappa^{CP} F_0^{CP}\big] \;-\; \gamma_y \big[-\alpha_S S_0 + \big(\alpha_F^{CP} + \tilde{\eta} S_0\big) F_0^{CP}\big] \;+\; \kappa^{DI} F_1^{DI} \;+\; \kappa^{CP} F_1^{CP} \end{aligned} \tag{48}$$

$$\theta_2 = \rho_\theta^2 \big(1 - \phi_S S_0\big)\big(1 - \phi_S S_1\big) \theta_0 \tag{49}$$

Interpretation. *Output gap.* Three mechanisms now shape y_2. First, the inherited gap y_1 propagates forward through ρ_1^{eff}—modulated by period-1 containment (hysteresis amplification via ψS_1) and period-1 CP (persistence reduction via $\eta_p F_1^{CP}$). Second, the period-1 flow terms contribute directly: the lockdown cost $-\alpha_S S_1$ and the contemporaneous CP effect $(\alpha_F^{CP} + \tilde{\eta} S_1) F_1^{CP}$. Third, the period-0 DI effect materializes: $\alpha_F^{DI} F_0^{DI}$, transmitted through the auxiliary state $z_1 = F_0^{DI}$. This is the first period in which DI contributes to output, confirming the two-period lag from deployment to effect: $F_0^{DI} \to z_1 \to y_2$.

Note that the period-0 CP (F_0^{CP}) continues to affect y_2 through two channels: its direct effect on y_1 (via $\alpha_F^{CP} + \tilde{\eta} S_0$) propagates forward at rate ρ_1^{eff}. This is a substantive difference from the single-channel specification, where F_0^{CP} had no output effect and therefore did not propagate.

Mortality. Period-2 mortality reflects the period-0 lockdown decision: $d_2 = \delta_\theta \rho_\theta (1 - \phi_S S_0) \theta_0$. This is the first period in which the planner's containment choice has a visible effect on mortality. The two-period lag $S_0 \to \theta_1 \to d_2$ is explicitly visible.

Debt. Rearranging (48) by period-0 controls:

$$b_2 = \underbrace{\gamma_y\,\alpha_S\,S_0}_{\substack{\text{Indirect lockdown}\\ \text{cost via } \gamma_y}} + \underbrace{(1+r)\,\kappa^{DI}\,F_0^{DI}}_{\substack{\text{Gross DI cost}\\ \text{(no output offset yet)}}}$$
$$+ \underbrace{\left[(1+r)\,\kappa^{CP} - \gamma_y\big(\alpha_F^{CP} + \tilde{\eta}\,S_0\big)\right] F_0^{CP}}_{\text{Net CP cost (with partial offset)}} + \underbrace{\kappa^{DI}\,F_1^{DI} + \kappa^{CP}\,F_1^{CP}}_{\text{Period-1 interventions}} \tag{50}$$

Two features distinguish this from the single-channel specification. First, DI deployed in period 0 has no output effect until y_2, so its net cost in b_2 is the gross cost $(1+r)\,\kappa^{DI}$ without any automatic stabilizer offset. The DI fiscal dividend appears one period later, in b_3. Second, CP now generates a partial fiscal dividend already in b_2: the term $-\gamma_y(\alpha_F^{CP}+\tilde{\eta}\,S_0)$ captures the automatic stabilizer savings from the improved y_1 attributable to period-0 CP. Under strict containment (S_0 large and $\tilde{\eta} < 0$), the cushioning channel reduces the offset—the fiscal dividend of CP is smaller precisely when the physical constraint on economic activity is tightest.

Period 2 → 3

Defining $\rho_2^{\text{eff}} \equiv \rho_y + \psi\,S_2 - \eta_p\,F_2^{CP}$:

Output gap.

$$y_3 = \rho_2^{\text{eff}}\,y_2 \;-\; \alpha_S\,S_2 \;+\; \big(\alpha_F^{CP} + \tilde{\eta}\,S_2\big)\,F_2^{CP} \;+\; \alpha_F^{DI}\,F_1^{DI} \tag{51}$$

Substituting (46) and defining the per-period flow contribution $\Phi_k \equiv -\alpha_S\,S_k + (\alpha_F^{CP} + \tilde{\eta}\,S_k)\,F_k^{CP} + \alpha_F^{DI}\,z_k$:

$$y_3 = \rho_2^{\text{eff}}\,\rho_1^{\text{eff}}\,\Phi_0 + \rho_2^{\text{eff}}\,\Phi_1 + \Phi_2 \tag{52}$$

This reveals the recursive structure: the output gap at any period $K+1$ is the sum of all past net flow contributions Φ_k, each attenuated by the product of effective persistence parameters from $k+1$ to K:

$$y_{K+1} = \sum_{k=0}^{K}\left[\prod_{j=k+1}^{K}\rho_j^{\text{eff}}\right]\Phi_k \tag{53}$$

CP enters this expression through two distinct mechanisms. First, it contributes to Φ_k through the level and cushioning channels $(\alpha_F^{CP} + \tilde{\eta}\,S_k)$, directly improving the output gap in the period of deployment. Second, it reduces the effective persistence ρ_j^{eff} through η_p, causing *all* past output gaps—whether generated by lockdowns, insufficient DI, or prior shocks—to dissipate faster. The first mechanism is contemporaneous and state-independent; the second is intertemporal and scales with $|y_k|$. It is the second mechanism

that makes CP increasingly valuable as the crisis deepens.

Mortality.

$$d_3 = \delta_\theta\, \theta_2 = \delta_\theta\, \rho_\theta^2 \big(1 - \phi_S\, S_0\big)\big(1 - \phi_S\, S_1\big)\, \theta_0 \tag{54}$$

Period-3 mortality depends on containment decisions in periods 0 and 1, each through the multiplicative suppression factor $(1 - \phi_S\, S_j)$. The period-2 lockdown S_2 does not yet affect d_3; it enters d_4. The product structure implies that lockdowns are complementary over time: the marginal effectiveness of S_1 depends on S_0.

Debt.

$$b_3 = (1 + r)\, b_2 \;-\; \gamma_y\, y_2 \;+\; \kappa^{DI}\, F_2^{DI} \;+\; \kappa^{CP}\, F_2^{CP} \tag{55}$$

Differentiating with respect to each period-0 control, holding period-1 and period-2 controls constant:

$$\begin{aligned} \frac{\partial b_3}{\partial S_0} &= (1 + r)\, \gamma_y\, \alpha_S \;+\; \gamma_y\, \rho_1^{\text{eff}}\, \alpha_S \\ &= \gamma_y\, \alpha_S \left[(1 + r) + \rho_1^{\text{eff}}\right] \end{aligned} \tag{56}$$

$$\frac{\partial b_3}{\partial F_0^{DI}} = (1 + r)^2\, \kappa^{DI} \;-\; \gamma_y\, \alpha_F^{DI} \tag{57}$$

$$\frac{\partial b_3}{\partial F_0^{CP}} = (1 + r)^2\, \kappa^{CP} \;-\; \gamma_y \big(\alpha_F^{CP} + \tilde{\eta}\, S_0\big) \left[(1 + r) + \rho_1^{\text{eff}}\right] \tag{58}$$

Interpretation. *Lockdown.* The lockdown has no direct debt cost but generates an indirect cost $\gamma_y\, \alpha_S\, [(1 + r) + \rho_1^{\text{eff}}]$ through the automatic stabilizer channel, compounding over two periods. The structure is identical to the single-channel specification because containment operates identically in both.

Demand injection. The fiscal dividend of period-0 DI materializes for the first time in b_3: the term $-\gamma_y\, \alpha_F^{DI}$ captures the automatic stabilizer savings from the improved y_2 attributable to F_0^{DI}. The net three-period cost is $(1 + r)^2\, \kappa^{DI} - \gamma_y\, \alpha_F^{DI}$—a fixed number independent of the state, reflecting the constant-multiplier property of DI. The dividend is delayed by one period relative to the single-channel specification (where DI affected output contemporaneously and generated a fiscal dividend in b_2).

Capacity preservation. Period-0 CP generates fiscal dividends through the automatic stabilizer offset $\gamma_y\, (\alpha_F^{CP} + \tilde{\eta}\, S_0)$, compounded by $[(1 + r) + \rho_1^{\text{eff}}]$ over two periods. This offset is state-dependent through S_0 (the cushioning channel modulates the output effect

that drives the dividend) and through ρ_1^{eff} (which governs how much of the improved y_1 propagates to y_2). The net three-period debt effect of CP becomes negative—i.e., CP becomes self-financing—when:

$$\left(\alpha_F^{CP} + \tilde{\eta}\, S_0\right)\left[(1+r) + \rho_1^{\text{eff}}\right] \;\geq\; \frac{(1+r)^2\,\kappa^{CP}}{\gamma_y} \tag{59}$$

This condition is satisfied when the combined level and cushioning effect is sufficiently large and when the effective persistence ρ_1^{eff} is high (i.e., the output gap is persistent enough for the automatic stabilizer savings to accumulate). The persistence-reduction channel (η_p) operates here indirectly: it enters through ρ_1^{eff} rather than through the direct output effect. A lower ρ_1^{eff} (achieved by high F_1^{CP}) reduces the cumulative fiscal dividend by dissipating the output gap faster—the paradoxical consequence that successful stabilization reduces the automatic stabilizer base.

D.8 Cumulative Effects of Period-0 Controls

Table 22 summarizes the effect of each period-0 control on all state variables across three periods. I define $\omega_0 \equiv \alpha_F^{CP} + \tilde{\eta}\, S_0$ as the net contemporaneous CP effect at the period-0 containment level.

Table 22: Period-0 Control Effects on State Variables (Three-Channel CP)

State	Control	Period 1	Period 2	Period 3
y	S_0	$-\alpha_S$	$-\rho_1^{\text{eff}}\,\alpha_S$	$-\rho_2^{\text{eff}}\,\rho_1^{\text{eff}}\,\alpha_S$
	F_0^{DI}	0	$+\alpha_F^{DI}$	$+\rho_2^{\text{eff}}\,\alpha_F^{DI}$
	F_0^{CP}	$+\omega_0$	$+\rho_1^{\text{eff}}\,\omega_0$	$+\rho_2^{\text{eff}}\,\rho_1^{\text{eff}}\,\omega_0$
d	S_0	0	$-\delta_\theta\,\rho_\theta\,\phi_S\,\theta_0$	via θ_2
	F_0^{DI}	0	0	0
	F_0^{CP}	0	0	0
b	S_0	0	$+\gamma_y\,\alpha_S$	Eq. (56)
	F_0^{DI}	$+\kappa^{DI}$	$(1{+}r)\,\kappa^{DI}$	$(1{+}r)^2\kappa^{DI}{-}\gamma_y\,\alpha_F^{DI}$
	F_0^{CP}	$+\kappa^{CP}$	$(1{+}r)\kappa^{CP}{-}\gamma_y\,\omega_0$	Eq. (58)
θ	S_0	$-\phi_S\,\rho_\theta\,\theta_0$	$-\phi_S\,\rho_\theta^2\,(1{-}\phi_S\,S_1)\,\theta_0$	cumulative
	F_0^{DI}	0	0	0
	F_0^{CP}	0	0	0

Notes: $\omega_0 \equiv \alpha_F^{CP} + \tilde{\eta}\, S_0$ is the net contemporaneous CP effect at the period-0 containment level; $\rho_k^{\text{eff}} \equiv \rho_y + \psi\, S_k - \eta_p\, F_k^{CP}$. The CP output effect decays at rate ρ_k^{eff}, with the persistence-reduction channel (η_p) entering through the attenuation factor rather than the flow term. The DI output effect first appears in period 2 (two-period lag) and decays at rate ρ_k^{eff} thereafter.

Comparison with single-channel specification. Three differences relative to the specification with only the persistence channel ($\eta\, F_k^{CP}\, y_k$) are noteworthy.

First, CP now has an immediate output effect from period 1 onward. In the single-channel specification, $\partial y_1/\partial F_0^{CP} = 0$ because the persistence channel requires a pre-existing output gap ($y_0 = 0$). In the three-channel specification, $\partial y_1/\partial F_0^{CP} = \omega_0 = \alpha_F^{CP} + \tilde{\eta}\, S_0 > 0$, operating through the level and cushioning channels.

Second, CP generates a fiscal dividend one period earlier. In the single-channel specification, the first non-trivial $\partial b/\partial F_0^{CP}$ net of the gross cost appears in b_3 (through the persistence channel's effect on y_2). In the three-channel specification, the automatic stabilizer offset $-\gamma_y\, \omega_0$ appears already in b_2.

Third, the DI fiscal dividend is delayed by one period. In the single-channel specification (with contemporaneous DI in the output equation), the DI offset $-\gamma_y\, \alpha_F^{DI}$ appears in b_2. In the augmented system, DI affects output only in y_2 (via $z_1 = F_0^{DI}$), so the offset appears in b_3. The economic content is identical—the timing difference reflects the explicit modeling of the disbursement lag.

Epidemiological dynamics.

$$\theta_3 = \rho_\theta^3 \prod_{j=0}^{2} \big(1 - \phi_S\, S_j\big)\, \theta_0 \tag{60}$$

The viral prevalence at period 3 is determined by the cumulative suppression effort across all three periods, each entering multiplicatively. This product structure implies that lockdowns are complementary over time: the marginal effectiveness of suppression in any period depends on the suppression in all other periods.

E iLQR Algorithm

The bilinear terms in (21) and (24) prevent the application of standard LQR. The iterative Linear-Quadratic Regulator (iLQR) of Li and Todorov (2004) handles this by successively linearizing around the current trajectory and solving the resulting time-varying LQR subproblem. Abhijeet and Chakravorty (2025) show that iLQR is equivalent to Sequential Quadratic Programming applied to the discrete-time optimal control problem. The notation follows Bertsekas (2017, Chapters 3–4).

E.1 Backward Pass: Q-Function Expansion

Let $(\bar{\boldsymbol{x}}_k^{(j)}, \bar{\boldsymbol{u}}_k^{(j)})_{k=0}^{N-1}$ denote the nominal trajectory at iteration j. Define deviations $\delta\boldsymbol{x}_k = \boldsymbol{x}_k - \bar{\boldsymbol{x}}_k^{(j)}$ and $\delta\boldsymbol{u}_k = \boldsymbol{u}_k - \bar{\boldsymbol{u}}_k^{(j)}$. Expanding the local action-value (Q-)function to second order:

$$Q(\delta\boldsymbol{x}_k, \delta\boldsymbol{u}_k) \approx \boldsymbol{Q}'_{x,k}\, \delta\boldsymbol{x}_k + \boldsymbol{Q}'_{u,k}\, \delta\boldsymbol{u}_k + \tfrac{1}{2}\delta\boldsymbol{x}'_k\, \boldsymbol{Q}_{xx,k}\, \delta\boldsymbol{x}_k + \delta\boldsymbol{u}'_k\, \boldsymbol{Q}_{ux,k}\, \delta\boldsymbol{x}_k + \tfrac{1}{2}\delta\boldsymbol{u}'_k\, \boldsymbol{Q}_{uu,k}\, \delta\boldsymbol{u}_k \tag{61}$$

where $\boldsymbol{p}_{k+1} \in \mathbb{R}^5$ and $\boldsymbol{P}_{k+1} \in \mathbb{R}^{5\times 5}$ are the gradient and Hessian of the cost-to-go at $k+1$, and the coefficients are:

$$\boldsymbol{Q}_{x,k} = \boldsymbol{Q}_k\, \bar{\boldsymbol{x}}_k^{(j)} + \tilde{\boldsymbol{A}}_k'\, \boldsymbol{p}_{k+1} \tag{62}$$

$$\boldsymbol{Q}_{u,k} = \boldsymbol{R}_k\, \bar{\boldsymbol{u}}_k^{(j)} + \tilde{\boldsymbol{B}}_k'\, \boldsymbol{p}_{k+1} + \boldsymbol{C}'\big(\boldsymbol{\lambda}_k + \boldsymbol{I}_\mu\, \boldsymbol{c}_k\big) \tag{63}$$

$$\boldsymbol{Q}_{xx,k} = \boldsymbol{Q}_k + \tilde{\boldsymbol{A}}_k'\, \boldsymbol{P}_{k+1}\, \tilde{\boldsymbol{A}}_k \tag{64}$$

$$\boldsymbol{Q}_{uu,k} = \boldsymbol{R}_k + \tilde{\boldsymbol{B}}_k'\, \boldsymbol{P}_{k+1}\, \tilde{\boldsymbol{B}}_k + \boldsymbol{C}'\, \boldsymbol{I}_\mu\, \boldsymbol{C} \tag{65}$$

$$\boldsymbol{Q}_{ux,k} = \tilde{\boldsymbol{B}}_k'\, \boldsymbol{P}_{k+1}\, \tilde{\boldsymbol{A}}_k \tag{66}$$

The terms involving $\boldsymbol{C}$, $\boldsymbol{\lambda}_k$, and $\boldsymbol{I}_\mu$ arise from the Augmented Lagrangian treatment of box constraints and are defined in Section E.4 below.

Minimizing (61) with respect to $\delta\boldsymbol{u}_k$ yields the feedforward correction $\boldsymbol{k}_k$ and the state-feedback gain $\boldsymbol{K}_k$:

$$\delta\boldsymbol{u}_k^* = \boldsymbol{k}_k + \boldsymbol{K}_k\, \delta\boldsymbol{x}_k, \qquad \boldsymbol{k}_k = -\boldsymbol{Q}_{uu,k}^{-1}\, \boldsymbol{Q}_{u,k}, \qquad \boldsymbol{K}_k = -\boldsymbol{Q}_{uu,k}^{-1}\, \boldsymbol{Q}_{ux,k} \tag{67}$$

Substituting back yields the updated cost-to-go:

$$\boldsymbol{p}_k = \boldsymbol{Q}_{x,k} + \boldsymbol{K}_k'\, \boldsymbol{Q}_{uu,k}\, \boldsymbol{k}_k \tag{68}$$

$$\boldsymbol{P}_k = \boldsymbol{Q}_{xx,k} - \boldsymbol{K}_k'\, \boldsymbol{Q}_{uu,k}\, \boldsymbol{K}_k \tag{69}$$

Equations (62)–(69) constitute the backward pass, initiated at $\boldsymbol{p}_N = \boldsymbol{Q}_N\, \bar{\boldsymbol{x}}_N^{(j)}$ and $\boldsymbol{P}_N = \boldsymbol{Q}_N$, solved backward from $k = N-1$ to $k = 0$.

E.2 Regularization

Positive definiteness of $\boldsymbol{Q}_{uu,k}$ is required for the gain computation (67). The explosive eigenvalues—$(1+r) > 1$ and $\rho_\theta(1-\phi_S\, \bar{S}_k) > 1$ in early pandemic periods—cause $\boldsymbol{P}_{k+1}$ to grow large, potentially rendering $\boldsymbol{Q}_{uu,k}$ ill-conditioned. Following **?**, Levenberg–Marquardt regularization replaces $\boldsymbol{Q}_{uu,k}$ with $\tilde{\boldsymbol{Q}}_{uu,k} = \boldsymbol{Q}_{uu,k} + \lambda\, \boldsymbol{I}_3$ in the gain computation, where $\lambda > 0$ is adapted: increased when $\boldsymbol{Q}_{uu,k}$ fails a Cholesky factorization check, reduced geometrically after each successful iteration.

E.3 Forward Pass

Given gains $(\boldsymbol{k}_k,\, \boldsymbol{K}_k)_{k=0}^{N-1}$ from the backward pass, the forward pass integrates the updated policy through the *nonlinear* dynamics with step-size $\alpha \in (0,\, 1]$:

$$\hat{\boldsymbol{u}}_k = \bar{\boldsymbol{u}}_k^{(j)} + \alpha\, \boldsymbol{k}_k + \boldsymbol{K}_k\big(\hat{\boldsymbol{x}}_k - \bar{\boldsymbol{x}}_k^{(j)}\big), \qquad \hat{\boldsymbol{u}}_k \leftarrow \mathrm{clip}(\hat{\boldsymbol{u}}_k,\, \boldsymbol{u}_{\min},\, \boldsymbol{u}_{\max}) \tag{70}$$

$$\hat{\boldsymbol{x}}_{k+1} = f(\hat{\boldsymbol{x}}_k,\, \hat{\boldsymbol{u}}_k,\, k) \tag{71}$$

starting from $\hat{\boldsymbol{x}}_0 = \boldsymbol{x}_0$. The step size α is initialized at unity and halved until the new trajectory achieves a sufficient decrease in total cost (Armijo condition). If no improvement is found within a predetermined number of halvings, λ is increased and the backward pass is repeated.

E.4 Box Constraints via Augmented Lagrangian

The control variables are subject to box constraints: $S_k \in [0,\, S_{\max}]$, $F_k^{DI} \in [0,\, \bar{F}^{DI}]$, $F_k^{CP} \in [0,\, \bar{F}^{CP}]$, where $S_{\max} = 0.86$ (empirical maximum) and $\bar{F}^{DI}$, $\bar{F}^{CP}$ are calibrated upper bounds. These are expressed as six inequality constraints $\boldsymbol{C}\,\boldsymbol{u}_k \le \boldsymbol{c}$ with:

$$\boldsymbol{C} = \begin{pmatrix} -1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \boldsymbol{c} = \begin{pmatrix} 0 \\ S_{\max} \\ 0 \\ \bar{F}^{DI} \\ 0 \\ \bar{F}^{CP} \end{pmatrix} \tag{72}$$

Rather than simple projection, constraints are enforced through an Augmented Lagrangian scheme (Howell et al., 2019). At each outer iteration, the Lagrange multipliers $\boldsymbol{\lambda}_k \in \mathbb{R}^6$ are updated: $\boldsymbol{\lambda}_k \leftarrow \max\big(\boldsymbol{0},\, \boldsymbol{\lambda}_k + \boldsymbol{\mu}_k \odot (\boldsymbol{C}\,\boldsymbol{u}_k - \boldsymbol{c})\big)$, where $\boldsymbol{\mu}_k$ are penalty parameters increased by a factor $\phi > 1$ after each outer iteration. The diagonal matrix $\boldsymbol{I}_\mu = \mathrm{diag}\big(\mu_k^{(i)} \cdot \mathbb{1}[c_i \text{ active}]\big)$ enters the Q-function coefficients (63) and (65), modifying the search direction to respect the constraints without requiring explicit projection during the backward pass.

E.5 Certainty Equivalence

At each iLQR iteration the linearized system is affine in the state and the objective is quadratic, satisfying the conditions for certainty equivalence (Bertsekas, 2017, Chapter 3.1). The optimal feedback gain $\boldsymbol{K}_k^{(j)}$ is therefore identical to the deterministic case ($\varepsilon_k \equiv 0$). Stochastic shocks increase the expected cost by $\sum_{k=0}^{N-1} \sigma^2\, \tilde{\boldsymbol{G}}'\, \boldsymbol{P}_{k+1}^{(j)}\, \tilde{\boldsymbol{G}}$—reflecting the irreducible cost of epidemiological uncertainty—but do not alter the optimal policy

rule.

E.6 Convergence

The algorithm terminates when the relative cost reduction falls below tolerance ϵ:

$$\frac{\left|J^{(j)} - J^{(j-1)}\right|}{\left|J^{(j)}\right|} < \epsilon, \qquad \epsilon = 10^{-6} \tag{73}$$

A maximum of 500 iterations is imposed as a safeguard; in practice, convergence is achieved within 40–80 inner iterations for all initializations tested. The bilinear terms render the problem non-convex, so the iLQR converges to a local optimum. To verify global optimality in the economically relevant region, I solve from 50 random initialization trajectories—drawn by perturbing the baseline control sequence uniformly within the feasible set—and confirm that all runs converge to the same cost J^* up to numerical tolerance. No alternative local optima are detected.

E.7 Parameters Full Table

Table 23: Variables, Parameters, and Calibration Status

Symbol	Description	Equation
State variables ($\boldsymbol{x}_k \in \mathbb{R}^4$, augmented to $\mathbb{R}^5$ with $z_k = F_{k-1}^{DI}$)		
y_k	Output gap (% deviation of real GDP from potential)	(??)
d_k	Excess mortality (deviation of death rate from seasonal average, per capita)	(??)
b_k	Public debt (deviation of debt-to-GDP ratio from pre-pandemic trend, pp)	(??)
θ_k	Infection rate (share of actively infected population)	(??)
z_k	Lagged demand injection ($z_k \equiv F_{k-1}^{DI}$)	(??)
Control variables ($\boldsymbol{u}_k \in \mathbb{R}^3$)		
S_k	Containment intensity (0: no restrictions, 1: full lockdown)	(??), (??)
F_k^{DI}	Demand injection (% of GDP)	(??), (??)
F_k^{CP}	Capacity preservation (% of GDP)	(??), (??)

Symbol	Description	Value	Source	Eq.
Epidemiological parameters (calibrated)				
ρ_θ	Effective reproduction rate (> 1: explosive)	1.25–1.60	Dhungel et al. (2022); Liu and Rocklöv (2021); Davies et al. (2021)	(??)
δ_θ	Infection fatality rate	0.005–0.025	COVID-19 Forecasting Team (2022b); Levin et al. (2020)	(??)
ϕ_S	NPI suppression effectiveness	0.4–0.7	Brauner et al. (2021); Haug et al. (2020)	(??)
Macroeconomic environment (calibrated)				
r	Real interest rate (quarterly)	≈ 0.001	OECD (2022)	(??)
β	Discount factor (quarterly)	0.99	Standard	(??)
c_H	Health expenditure per unit infection	0.02	International Monetary Fund (2022); OECD (2023)	(??)
Output dynamics (estimated, Section 4)				
ρ_y	Baseline output persistence	—		(??)
ψ	Lockdown-induced hysteresis (> 0)	—		(??)
α_S	Contractionary effect of NPIs	—		(??)
α_F^{CP}	CP baseline contemporaneous effect	—		(??)
$\tilde{\eta}$	CP–containment interaction (< 0: lockdown constraint)	—		(??)
η_p	CP structural preservation effect	—		(??)
α_F^{DI}	DI multiplier (lagged)	—		(??)
β_d	Behavioral fear channel (< 0)	—		(??)
Debt dynamics (estimated, Section 4)				
γ_y	Budget semi-elasticity (automatic stabilizers)	—		(??)
κ_F^{CP}	Budgetary cost of CP	—		(??)
κ_F^{DI}	Budgetary cost of DI	—		(??)
Objective function parameters				
w_y, w_d, w_b	Running cost weights (output, mortality, debt)			(??)
W_b	Terminal debt cost			(??)
r_S, r_F^{DI}, r_F^{CP}	Control implementation costs			(??)

Table 24: Calibrated Parameters

Parameter	Description	Value	Source
Epidemiological dynamics (wave-specific)			
ρ_θ^{W1}	Eff. reproduction rate, Ancestral (Q1–Q2 2020)	1.30	Flaxman et al. (2020); Dhungel et al.
ρ_θ^{W2}	Eff. reproduction rate, 2nd wave (Q3–Q4 2020)	1.25	Flaxman et al. (2020)
ρ_θ^{W3}	Eff. reproduction rate, Alpha (Q1–Q2 2021)	1.45	Davies et al. (2021); Liu and Rocklöv
ρ_θ^{W4}	Eff. reproduction rate, Delta (Q3 2021)	1.55	Liu and Rocklöv (2021)
ρ_θ^{W5}	Eff. reproduction rate, Omicron (Q4 2021)	1.70	Liu and Rocklöv (2021)
δ_θ^{W1}	Infection fatality rate, Ancestral	0.020	COVID-19 Forecasting Team (2022b)
δ_θ^{W2}	Infection fatality rate, 2nd wave	0.015	COVID-19 Forecasting Team (2022b)
δ_θ^{W3}	Infection fatality rate, Alpha	0.012	COVID-19 Forecasting Team (2022b)
δ_θ^{W4}	Infection fatality rate, Delta	0.008	COVID-19 Forecasting Team (2022b)
δ_θ^{W5}	Infection fatality rate, Omicron	0.003	COVID-19 Forecasting Team (2022b)
ϕ_S	NPI suppression effectiveness	0.55	Brauner et al. (2021); Haug et al. (20
Macroeconomic environment			
r	Real interest rate (quarterly)	0.001	OECD (2022)
β	Discount factor (quarterly)	0.99	Standard
c_H	Health expenditure per unit infection	0.02	International Monetary Fund (2022);

F Correlation Matrix and Variation Decomposition

Table 25: Within vs. Between Variation Decomposition

Variable	Overall SD	Between SD	Within SD	Within (%)	Between (%)
y_{ik} (pp of potential)	4.766	3.018	3.688	77.4	63.3
S_{ik} (0–100)	18.696	6.143	17.658	94.4	32.9
F_{ik}^{CP} (pp GDP)	3.032	0.691	2.953	97.4	22.8
F_{ik}^{DI} (pp GDP)	0.705	0.251	0.659	93.5	35.6
F_{ik}^{H} (pp GDP)	0.381	0.123	0.361	94.7	32.3
d_{ik} (P-score, %)	14.873	7.603	12.783	85.9	51.1
θ_{ik} (% of pop.)	1.889	0.533	1.813	95.9	28.2
Vax_{ik} (% of pop.)	33.038	4.212	32.768	99.2	12.7

Notes: 38 OECD countries, Q1 2020–Q2 2022 ($N = 380$). Overall SD: total standard deviation. Between SD: standard deviation of country means. Within SD: standard deviation of deviations from country means. Within (%) = Within SD / Overall SD × 100. Country fixed effects absorb between-variation; within-variation identifies the coefficients. All policy variables exhibit within-shares above 93%, confirming that the identifying variation survives absorption of country fixed effects.

Table 26: Pairwise Correlation Matrix

	y_{ik}	S_{ik}	F_{ik}^{CP}	F_{ik}^{DI}	d_{ik}	θ_{ik}	Vax_{ik}
y_{ik}	1						
S_{ik}	−0.43	1					
F_{ik}^{CP}	−0.25	0.13	1				
F_{ik}^{DI}	−0.11	0.12	0.25	1			
d_{ik}	−0.09	0.20	−0.15	−0.04	1		
θ_{ik}	0.20	−0.39	−0.19	−0.17	0.32	1	
Vax_{ik}	0.43	−0.56	−0.36	−0.27	0.04	0.53	1

Notes: Pearson correlations, estimation sample (38 countries, Q1 2020–Q2 2022, $N = 380$). Lower triangle reported. The correlations $r(S, F^{CP}) = 0.13$ and $r(S, F^{DI}) = 0.12$ confirm that fiscal composition is only weakly correlated with containment intensity—the key identifying condition for the $S \times F^{CP}$ interaction term.